



(DEEMED TO BE UNIVERSITY)

FRONT END DEVELOPMENT FRAMEWORKS AND UI ENGINEERING

25CS1201E

Real Time Problems there Titles

400 Real-World Problems × 8 Industry Domains

HTML • CSS • JavaScript • React.js • Node.js • npm • Postman • MySQL

Document Overview

This document contains 400 complete hackathon-grade frontend engineering problems across 8 industry domains. Each problem includes a description, tools & technologies required, and 10 implementation modules covering real-world scenarios.

Domain	Problems	Core Focus
✈ Airlines Domain	50	UI/UX, APIs, Auth, Admin, Testing
🏥 Healthcare Domain	50	UI/UX, APIs, Auth, Admin, Testing
🎓 Education Domain	50	UI/UX, APIs, Auth, Admin, Testing
🏨 Hotels Domain	50	UI/UX, APIs, Auth, Admin, Testing
🎓 College Domain	50	UI/UX, APIs, Auth, Admin, Testing
🏦 Bank Domain	50	UI/UX, APIs, Auth, Admin, Testing
🛒 Online Shopping Apps Domain	50	UI/UX, APIs, Auth, Admin, Testing
🏛 Meeseva (Gov Services) Domain	50	UI/UX, APIs, Auth, Admin, Testing

Universal Technology Stack

All problems in this document are designed to be solved using the following unified technology stack:

Frontend: HTML5, CSS3, JavaScript (ES6+), React.js (Hooks, Router, Context API / Redux), Bootstrap / Tailwind CSS

Backend: Node.js, Express.js, npm package management, REST API development

API Testing: Postman (Collections, Environments, Automated Test Suites, Newman CLI)

Database: MySQL (Optional — for persistence; problems are completable without DB), LocalStorage for client-state

Libraries: Axios (HTTP), React Router DOM, Chart.js / Recharts (data viz), React Hook Form, Moment.js / Day.js

Dev Tools: VS Code, Git, Chrome DevTools, ESLint, Prettier, npm scripts

Deployment: Localhost / Vercel / Netlify (optional for demo)

✈ Airlines Domain

This section contains 50 complete hackathon problem statements for the Airlines Domain. Each problem is ready for immediate prototype development.

Problem 1: Flight Search & Booking Engine

Domain: Airlines

Difficulty: ★ Beginner

Description:

Allow users to search flights by source, destination, date, class; display real-time results with filters for price, airline, stops.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Flight search, Seat selection, Booking summary, Payment gateway mock, E-ticket generation, Booking history, Filters & sorting, User auth, Notifications, Admin dashboard

Problem 2: Seat Selection System

Domain: Airlines

Difficulty: ★ ★ Intermediate

Description:

Interactive seat map with real-time availability, colour-coded by class (Economy, Business, First), blocked seats, and price breakdown per seat.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Seat map renderer, Class toggle, Availability API, Seat lock timer, Price calculator, Booking confirm, User profile, Responsive layout, Seat legend, Accessibility mode

Problem 3: Online Check-In Portal

Domain: Airlines

Difficulty: ★ ★ ★ Advanced

Description:

Web-based check-in allowing passengers to select seats, add baggage, choose meals, download boarding passes as PDF.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Check-in flow, Baggage add-on, Meal preference, Boarding pass PDF, QR code generator, Passenger details form, Route display, Notifications, Multi-pax support, Admin view

Problem 4: Flight Status Tracker

Domain: Airlines

Difficulty: ★★ Intermediate

Description:

Real-time flight status updates with departure/arrival times, gate info, delays, and push notifications for subscribed flights.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Status dashboard, Live updates (polling), Push alerts, Search by flight no., Map view, History log, Notification subscription, Filter by airline, Mobile responsive, Admin panel

Problem 5: Baggage Tracking System

Domain: Airlines

Difficulty: ★★★ Advanced

Description:

Track checked-in baggage from check-in to carousel with live status updates, lost baggage reporting, and claim filing.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Bag ID entry, Status timeline, Live updates, Lost report form, Claim upload, Notification email mock, Admin status update, Multi-bag support, History view, FAQ module

Problem 6: Airline Loyalty Program Portal

Domain: Airlines

Difficulty: ★ Beginner

Description:

Manage frequent flyer points, tier status, redemption options, transfer to partners, and view point expiry dates.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Points dashboard, Tier display, Earn history, Redeem flow, Partner transfer, Expiry alert, Profile settings, Statement download, Referral system, Admin panel

Problem 7: Flight Delay Compensation Claim

Domain: Airlines

Difficulty: ★★ Intermediate

Description:

Submit and track EU261/DOT compensation claims for delayed or cancelled flights with document upload and status tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Claim form, Document upload, Eligibility checker, Claim status, Email notification, History log, FAQ, Admin review, Export PDF, Support chat mock

Problem 8: Airport Lounge Access Manager

Domain: Airlines

Difficulty: ★★★ Advanced

Description:

Book and manage lounge access with membership verification, guest passes, amenities display, and entry time slots.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Lounge search, Booking form, Membership check, Guest pass, Slot picker, QR entry pass, Amenities display, History, Notifications, Admin dashboard

Problem 9: Multi-City Trip Planner

Domain: Airlines

Difficulty: ★ Beginner

Description:

Plan complex multi-city itineraries with layover management, cost aggregation, visa info, and exportable travel summary.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Route builder, Layover display, Cost summary, Visa info panel, Date picker, Trip export PDF, Saved trips, Share itinerary, Airline compare, Map view

Problem 10: Airline Crew Scheduling UI

Domain: Airlines

Difficulty: ★★ Intermediate

Description:

Crew management dashboard for scheduling pilots and cabin crew, tracking duty hours, leave requests, and regulatory compliance.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Roster view, Duty hour tracker, Leave request form, Conflict alerts, Assignment drag-drop, Compliance check, Profile management, Shift export, Notifications, Admin panel

Problem 11: Aircraft Maintenance Tracker

Domain: Airlines

Difficulty: ★ Beginner

Description:

Log and track scheduled and unscheduled aircraft maintenance tasks, parts inventory, and technician assignments.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Task list, Schedule calendar, Technician assign, Parts inventory, Status update, Alert system, Maintenance history, Export report, Photo upload, Admin panel

Problem 12: Passenger Feedback System

Domain: Airlines

Difficulty: ★ ★ Intermediate

Description:

Post-flight feedback portal for rating cabin crew, food, cleanliness, punctuality with sentiment-scored responses.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Rating form, Category scoring, Comments, Submission confirm, Feedback history, Sentiment display, Analytics chart, Email trigger, Admin review, Export CSV

Problem 13: Lost & Found Portal

Domain: Airlines

Difficulty: ★ ★ ★ Advanced

Description:

Report and track lost items on flights or airports, match with found items in system, arrange collection or courier.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Lost report form, Found item list, Match engine, Status tracker, Photo upload, Collection form, Email notify, Admin match, History, FAQ

Problem 14: Dynamic Pricing Dashboard

Domain: Airlines

Difficulty: ★ ★ Intermediate

Description:

Visualise fare fluctuations over time, compare competitor pricing, set pricing rules, and predict optimal fare windows.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Price chart, Competitor compare, Rule builder, Forecast display, Date range picker, Alerts, Export data, Route filter, History view, Admin panel

Problem 15: Group Booking Management

Domain: Airlines

Difficulty: ★ ★ ★ Advanced

Description:

Handle group travel bookings (10+ passengers) with split payments, group seat allocation, and one-click bulk changes.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Group form, Passenger list, Seat allocate, Split payment mock, Bulk edit, Booking summary, Email all, Invoice generate, History, Admin panel

Problem 16: Travel Insurance Add-On Flow

Domain: Airlines

Difficulty: ★ Beginner

Description:

Integrate travel insurance selection during booking with coverage comparison, premium calculator, and instant policy issuance.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Insurance picker, Coverage compare, Premium calc, Policy form, Payment mock, Policy PDF, Profile link, History, Admin panel, FAQ

Problem 17: Visa & Travel Requirements Checker

Domain: Airlines

Difficulty: ★★ Intermediate

Description:

Check visa requirements, health advisories, and entry rules by nationality and destination with real-time government data.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Country picker, Nationality select, Requirement display, Health advisory, Update date, Bookmark country, Share feature, Search, FAQ, Admin update panel

Problem 18: Airline Carbon Offset Calculator

Domain: Airlines

Difficulty: ★★★ Advanced

Description:

Calculate CO2 emissions per flight route, offer offset purchase options, and display sustainability impact dashboard.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Route input, Emission calc, Offset packages, Purchase mock, Impact display, Certificate PDF, History, Leaderboard, Admin panel, Share widget

Problem 19: In-Flight Entertainment Catalogue

Domain: Airlines

Difficulty: ★ Beginner

Description:

Browse and pre-select movies, music, and games for an upcoming flight; sync preferences with seat system.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Catalogue grid, Category filter, Search, Favourites, Pre-select flow, Sync mock, Preview player, History, Profile link, Admin upload

Problem 20: Ancillary Services Marketplace

Domain: Airlines

Difficulty: ★★ Intermediate

Description:

Upsell add-ons (extra baggage, priority boarding, seat upgrades, lounge access) during or after booking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Add-on catalogue, Bundle builder, Price comparison, Cart, Payment mock, Order confirm, Email receipt, Profile link, History, Admin panel

Problem 21: Airport Wayfinding App UI

Domain: Airlines

Difficulty: ★ Beginner

Description:

Interactive airport map for navigating terminals, finding gates, lounges, restaurants, security queues, and parking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Map renderer, Search POI, Route highlight, Floor selector, Gate info, Live queue mock, Favourites, QR scan mock, Accessibility toggle, Admin update

Problem 22: Flight Comparison Tool

Domain: Airlines

Difficulty: ★★ Intermediate

Description:

Compare flights across dates and airlines with price calendars, baggage policy comparison, and on-time performance stats.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Date price calendar, Airline compare, Baggage policy grid, OTP stats, Route map, Filter panel, Share link, Sort options, Mobile view, Export

Problem 23: Pilot Training Records Portal

Domain: Airlines

Difficulty: ★★★ Advanced

Description:

Maintain pilot training logs, certifications, simulator hours, medical checks, and regulatory renewal reminders.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Training log, Cert tracker, Hours chart, Medical check list, Renewal alerts, Upload docs, Profile, Export PDF, Admin panel, History

Problem 24: Airline API Testing Dashboard

Domain: Airlines

Difficulty: ★ ★ Intermediate

Description:

Postman-style dashboard to test airline REST APIs (search, book, cancel) with request builder, response viewer, and test suites.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Request builder, Auth header, Response viewer, Test suite, Environment vars, History, Mock server, Export results, Error logger, Docs panel

Problem 25: Revenue Management Analytics UI

Domain: Airlines

Difficulty: ★ ★ ★ Advanced

Description:

Visualise load factors, yield per route, ancillary revenue, and forecast accuracy with interactive charts and drill-downs.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Dashboard KPIs, Load factor chart, Yield chart, Route drill-down, Date range, Forecast vs actual, Export, Alert thresholds, Compare periods, Admin panel

Problem 26: Passenger Name Record (PNR) Manager

Domain: Airlines

Difficulty: ★ Beginner

Description:

Search, view, modify, and cancel PNRs with full booking details, SSR codes, and audit trail.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

PNR search, Detail view, Modify form, Cancel flow, SSR manage, Audit log, Print/export, Notifications, Admin panel, History

Problem 27: Airline Social Media Monitor

Domain: Airlines

Difficulty: ★★ Intermediate

Description:

Track airline brand mentions, complaints, and compliments across platforms with sentiment score and response workflow.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Feed aggregator mock, Sentiment badge, Category filter, Response template, Assign agent, Status track, Analytics, Export, Notification, Admin panel

Problem 28: Flight Training Simulator Booking

Domain: Airlines

Difficulty: ★★★ Advanced

Description:

Book simulator slots for pilot training, manage recurrent training schedules, and track simulator utilisation.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Slot calendar, Book form, Utilisation chart, Recurrent reminder, Instructor assign, Confirm email mock, History, Admin panel, Export, Notifications

Problem 29: Airport Retail & F&B Pre-Order

Domain: Airlines

Difficulty: ★ Beginner

Description:

Pre-order food, duty-free, and retail items for collection at airport by flight time, with order tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Catalogue, Cart, Flight link, Time slot, Order confirm, Order track, Payment mock, Collection QR, History, Admin panel

Problem 30: Airline Refund & Cancellation Portal

Domain: Airlines

Difficulty: ★★ Intermediate

Description:

Self-service flight cancellation with fare rule display, refund eligibility check, processing timeline, and refund status.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Cancel flow, Fare rules display, Eligibility check, Refund calc, Submit form, Status tracker, Email notify, History, FAQ, Admin panel

Problem 31: Charter Flight Request System

Domain: Airlines

Difficulty: ★ Beginner

Description:

Request and manage charter flights with route, pax count, catering, and special requirements submission.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Request form, Route map, Pax config, Catering options, Requirement notes, Quote display mock, Status track, Email confirm, History, Admin panel

Problem 32: Airline HR & Onboarding Portal

Domain: Airlines

Difficulty: ★★ Intermediate

Description:

Onboard new airline staff with document submission, training assignment, badge request, and HR task checklist.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Onboard form, Doc upload, Training list, Badge request, HR checklist, Progress tracker, Email notify, Profile, Admin panel, Reports

Problem 33: Airport Parking Reservation

Domain: Airlines

Difficulty: ★★★ Advanced

Description:

Book airport parking spots with terminal proximity, vehicle details, pricing comparison, and entry/exit barcode.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Spot search, Booking form, Vehicle input, Price compare, Barcode generate, Payment mock, Email confirm, History, Modify/cancel, Admin panel

Problem 34: Airline Disruption Management UI

Domain: Airlines

Difficulty: ★★ Intermediate

Description:

Operations centre UI for managing flight disruptions — re-accommodation, hotel vouchers, meal vouchers, and passenger communication.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Disruption dashboard, Pax re-accom, Hotel voucher, Meal voucher, Comm send mock, Status board, Priority queue, Export, History, Admin panel

Problem 35: Frequent Flyer Enrolment Flow

Domain: Airlines

Difficulty: ★★★ Advanced

Description:

Multi-step enrolment wizard for frequent flyer program with tier preview, benefits explainer, and instant number issuance.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Multi-step form, Benefits display, Tier preview, ID upload, Confirm screen, Welcome email mock, Profile link, History, Share widget, Admin panel

Problem 36: Airline Sustainability Dashboard

Domain: Airlines

Difficulty: ★ Beginner

Description:

Track and report airline sustainability KPIs — fuel efficiency, carbon per seat, offset purchased, SAF usage, noise reduction.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

KPI cards, Trend charts, Route breakdown, SAF tracker, Offset record, Export PDF, Alert thresholds, Compare periods, Admin panel, Public view toggle

Problem 37: Cargo Booking & Tracking System

Domain: Airlines

Difficulty: ★★ Intermediate

Description:

Book cargo space on flights, input shipment details, generate AWB, and track shipment status from origin to destination.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Booking form, Weight/vol calc, AWB generator, Status tracker, Route display, Document upload, Email notify, History, Admin panel, Postman testing

Problem 38: Airline Corporate Travel Portal

Domain: Airlines

Difficulty: ★★★ Advanced

Description:

Corporate clients manage employee travel, set policy rules, approve trips, track spend, and generate travel reports.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Employee list, Trip request, Approval flow, Policy rules, Spend dashboard, Report export, Profile, Notifications, Admin panel, Postman API test

Problem 39: Aircraft Configuration Designer

Domain: Airlines

Difficulty: ★ Beginner

Description:

Design cabin layouts with drag-and-drop seat configuration, class zones, galley/lavatory placement, and capacity summary.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Drag-drop canvas, Seat types, Class zone, Galley/lav place, Capacity calc, Save config, Export image, Compare configs, History, Admin panel

Problem 40: Flight Bidding & Upgrade Auction

Domain: Airlines

Difficulty: ★★ Intermediate

Description:

Allow passengers to bid for seat upgrades; airline side manages auction parameters, auto-award, and notification.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Bid form, Auction timer, Current bid display, Auto-award logic, Notify winner, Flight filter, Bid history, Payment mock, Admin panel, Postman test

Problem 41: Airline Mobile Wallet Integration

Domain: Airlines

Difficulty: ★ Beginner

Description:

Store boarding passes, vouchers, and loyalty cards in a digital wallet UI with expiry alerts and barcode display.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Wallet dashboard, Add pass, Barcode display, Expiry alert, Share pass, Delete pass, History, Notification, Profile, Admin panel

Problem 42: Tarmac Delay Management UI

Domain: Airlines

Difficulty: ★★ Intermediate

Description:

Track and manage tarmac delay incidents with passenger communication, catering requests, deplane decision support.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Delay log, Timer display, Pax count, Catering request, Deplane decision, Comm send mock, Regulatory timer alert, History, Export, Admin panel

Problem 43: Airline Vendor Management Portal

Domain: Airlines

Difficulty: ★★★ Advanced

Description:

Manage catering, cleaning, and ground handling vendors with SLA tracking, invoice processing, and performance ratings.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Vendor list, Profile view, Contract upload, SLA tracker, Invoice submit, Rating form, Issue log, Notifications, Export, Admin panel

Problem 44: Flight Academy Admission Portal

Domain: Airlines

Difficulty: ★ ★ Intermediate

Description:

Apply to airline training academy with eligibility check, document upload, test scheduling, and admission status tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Eligibility form, Doc upload, Test scheduler, Status tracker, Score display, Offer letter mock, FAQ, Profile, Email notify, Admin panel

Problem 45: Airline Ops Briefing Dashboard

Domain: Airlines

Difficulty: ★ ★ ★ Advanced

Description:

Daily operational briefing for airport managers covering slot utilisation, delays, staffing, weather, and gate conflicts.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

KPI summary, Slot chart, Delay heatmap, Staff table, Weather widget mock, Gate conflict alert, Export brief, Date picker, Notifications, Admin panel

Problem 46: Passenger Accessibility Services

Domain: Airlines

Difficulty: ★ Beginner

Description:

Request and track special assistance (wheelchair, visual impairment, dietary) with assignment to ground staff.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Assistance form, Service types, Status track, Staff assign, Email notify, Flight link, History, Admin panel, Postman test, FAQ

Problem 47: Interline Baggage Claim Portal

Domain: Airlines

Difficulty: ★ ★ Intermediate

Description:

Handle interline baggage claims between partner airlines with cross-reference of booking codes and baggage tags.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Claim form, Bag tag input, Partner airline select, Status display, Document upload, Email notify, History, Admin review, Export, FAQ

Problem 48: Airline News & Press Room

Domain: Airlines

Difficulty: ★ ★ ★ Advanced

Description:

Manage and publish press releases, media kits, investor news, and route announcements with searchable archive.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

News list, Article view, Category filter, Search, Media kit download, Subscribe mock, Admin publish, Rich text editor, Date filter, Export

Problem 49: Ground Crew Coordination App

Domain: Airlines

Difficulty: ★ Beginner

Description:

Real-time coordination between ground crew (fuelling, catering, cleaning, baggage) with task assignments and completion checks.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Task board, Crew assign, Status update, Flight link, Timer, Notification, Checklist, History, Export report, Admin panel



Healthcare Domain

This section contains 50 complete hackathon problem statements for the Healthcare Domain. Each problem is ready for immediate prototype development.

Problem 50: Patient Registration & EMR Portal

Domain: Healthcare

Difficulty: ★ Beginner

Description:

Digital patient onboarding with medical history, allergies, current medications, emergency contacts, and consent forms.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Registration form, Medical history, Allergy input, Medication list, Emergency contact, Consent form, Profile photo, Insurance details, Admin panel, Postman API

Problem 51: Doctor Appointment Booking System

Domain: Healthcare

Difficulty: ★ ★ Intermediate

Description:

Search doctors by specialty/location, view availability calendars, book/reschedule/cancel appointments with reminders.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Doctor search, Specialty filter, Calendar view, Slot picker, Booking confirm, Reminders, Booking history, Cancel/reschedule, Patient profile, Admin panel

Problem 52: Telemedicine Video Consultation UI

Domain: Healthcare

Difficulty: ★ ★ ★ Advanced

Description:

Video call interface for doctor-patient consultations with pre-call symptom check, prescription generation, and post-call notes.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Pre-call check, Video interface mock, Prescription form, Symptom input, Doctor notes, PDF prescription, Post-call summary, History, Notifications, Admin

Problem 53: Prescription Management System

Domain: Healthcare

Difficulty: ★★ Intermediate

Description:

Manage digital prescriptions with drug database, dosage validation, refill requests, pharmacy routing, and adherence tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Prescription form, Drug search, Dosage validator, Refill request, Pharmacy assign, PDF export, Adherence tracker, History, Notifications, Admin panel

Problem 54: Hospital Bed Management Dashboard

Domain: Healthcare

Difficulty: ★★★ Advanced

Description:

Real-time bed availability across wards with admissions, discharges, transfers, maintenance status, and capacity alerts.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Bed map, Ward filter, Admit/discharge form, Transfer flow, Maintenance flag, Capacity chart, Alert system, History, Export, Admin panel

Problem 55: Lab Test Ordering & Results Portal

Domain: Healthcare

Difficulty: ★ Beginner

Description:

Order lab tests, track sample status, receive results with reference ranges, flagged abnormalities, and doctor review.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Test catalogue, Order form, Sample status track, Result display, Reference range, Flag abnormal, Doctor review, PDF download, History, Admin

Problem 56: Medical Imaging Viewer UI

Domain: Healthcare

Difficulty: ★★ Intermediate

Description:

Upload and view DICOM-like medical images (X-ray, MRI, CT) with annotations, zoom, measurement tools, and report attachment.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Image upload, Viewer canvas, Zoom/pan, Annotation tool, Measurement, Report attach, Share link, History, Export, Admin panel

Problem 57: Patient Discharge Summary Generator

Domain: Healthcare

Difficulty: ★★★ Advanced

Description:

Auto-generate discharge summaries from structured data (diagnosis, procedures, medications, follow-up) in printable format.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Data input form, Template engine, Diagnosis coding, Procedure list, Medication summary, Follow-up plan, PDF generate, History, Signature mock, Admin

Problem 58: Emergency Triage System

Domain: Healthcare

Difficulty: ★ Beginner

Description:

Colour-coded triage interface for A&E departments to assign severity levels, track wait times, and alert for critical cases.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Triage form, Severity badge, Wait timer, Priority queue, Alert system, Bed link, Status update, History, Export, Admin panel

Problem 59: Pharmacy Inventory Management

Domain: Healthcare

Difficulty: ★★ Intermediate

Description:

Track drug stock levels, expiry dates, reorder alerts, supplier orders, and dispensing records with barcode lookup.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Stock list, Expiry alert, Reorder form, Supplier orders, Dispense log, Barcode lookup, Chart analytics, Export, Notifications, Admin panel

Problem 60: Clinical Trial Recruitment Portal

Domain: Healthcare

Difficulty: ★ Beginner

Description:

Enrol eligible patients into clinical trials with eligibility screening, consent management, and randomisation assignment.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Eligibility form, Consent upload, Trial list, Assignment display, Status tracker, Email notify, History, Admin review, Export, FAQ

Problem 61: Health Insurance Claims Portal

Domain: Healthcare

Difficulty: ★ ★ Intermediate

Description:

Submit and track insurance claims with document upload, eligibility verification, claim status, and payment history.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Claim form, Doc upload, Eligibility check, Status tracker, Payment history, Appeal form, PDF export, Notifications, Admin panel, Postman API

Problem 62: Nurse Shift Scheduling System

Domain: Healthcare

Difficulty: ★ ★ ★ Advanced

Description:

Schedule nurses across wards and shifts with skill-matching, overtime tracking, leave management, and conflict alerts.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Roster builder, Shift assign, Skill match, Overtime tracker, Leave request, Conflict alert, Export schedule, Notifications, Profile, Admin panel

Problem 63: Chronic Disease Monitoring Dashboard

Domain: Healthcare

Difficulty: ★★ Intermediate

Description:

Monitor patients with chronic conditions (diabetes, hypertension) with vitals logging, trend charts, and threshold alerts.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Vitals log form, Trend charts, Threshold alerts, Medication tracker, Doctor notes, Export PDF, History, Profile, Notification, Admin panel

Problem 64: Mental Health Self-Assessment Tool

Domain: Healthcare

Difficulty: ★★★ Advanced

Description:

Validated mental health screeners (PHQ-9, GAD-7) with scoring, recommendations, and referral pathway to professionals.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Questionnaire UI, Scoring engine, Result display, Recommendation, Referral form, History, Anonymise option, Resources page, Admin panel, Export

Problem 65: Vaccination Record & Scheduler

Domain: Healthcare

Difficulty: ★ Beginner

Description:

Maintain immunisation history, schedule upcoming vaccines by age/risk, send reminders, and generate vaccination certificates.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Record list, Vaccine scheduler, Reminder system, Certificate PDF, Upload past records, Age-based planner, History, QR code mock, Admin panel, Notifications

Problem 66: ICU Patient Monitor Interface

Domain: Healthcare

Difficulty: ★★ Intermediate

Description:

Simulated ICU monitoring dashboard displaying vitals (HR, BP, SpO2, temp) with alarm configuration and nurse call trigger.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Vitals display, Trend chart, Alarm config, Nurse call button, Status badge, Patient profile, History log, Export data, Notification, Admin panel

Problem 67: Medical Supply Chain Tracker

Domain: Healthcare

Difficulty: ★★★ Advanced

Description:

Track medical supplies from vendor to ward with purchase orders, delivery status, quality check, and usage analytics.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

PO form, Delivery track, Quality check, Usage chart, Alert low stock, Vendor list, Export, Notifications, Admin panel, Postman API

Problem 68: Patient Portal (MyHealth)

Domain: Healthcare

Difficulty: ★ Beginner

Description:

Self-service patient portal for viewing records, test results, upcoming appointments, billing, and secure messaging to doctors.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Dashboard, Records view, Results display, Appointment list, Billing summary, Secure message, Profile, Notification, Export, Admin panel

Problem 69: Radiology Report Turnaround Tracker

Domain: Healthcare

Difficulty: ★★ Intermediate

Description:

Track radiology report turnaround times by modality and radiologist, with SLA alerts, workload balancing, and trend analysis.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Report queue, Status update, TAT chart, Radiologist assign, SLA alert, Filter panel, Export, History, Notifications, Admin panel

Problem 70: Hospital Staff Directory & Paging

Domain: Healthcare

Difficulty: ★ Beginner

Description:

Searchable staff directory with on-call schedules, department paging, and direct message mock for clinical coordination.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Staff search, Department filter, On-call display, Page form, Message mock, Profile view, Schedule link, Export, Notifications, Admin

Problem 71: Consent Management System

Domain: Healthcare

Difficulty: ★★ Intermediate

Description:

Manage patient consents digitally — procedure consents, data sharing, research participation — with version control and e-signature.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Consent list, Form builder, E-signature mock, Version history, Patient link, Doctor assign, PDF export, Status tracker, Notifications, Admin

Problem 72: Dietary & Nutrition Planner

Domain: Healthcare

Difficulty: ★★★ Advanced

Description:

Inpatient meal planning based on dietary restrictions, allergies, medical conditions, and caloric requirements with kitchen orders.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Patient dietary profile, Meal planner, Allergy check, Calorie calc, Kitchen order, Approval flow, History, Export, Notifications, Admin panel

Problem 73: Wound Care Tracking System

Domain: Healthcare

Difficulty: ★ ★ Intermediate

Description:

Document wound assessments with photo uploads, measurement inputs, healing progress charts, and care plan management.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Assessment form, Photo upload, Measurement input, Healing chart, Care plan, Alerts, Assign nurse, History, Export PDF, Admin panel

Problem 74: Hospital Billing & Invoice System

Domain: Healthcare

Difficulty: ★ ★ ★ Advanced

Description:

Generate and manage patient bills with itemised charges, insurance deductions, payment recording, and receipt generation.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Bill summary, Itemised list, Insurance deduct, Payment form, Receipt PDF, Dispute form, History, Export, Notifications, Admin panel

Problem 75: Patient Transfer Coordination

Domain: Healthcare

Difficulty: ★ Beginner

Description:

Coordinate intra-hospital and inter-hospital patient transfers with bed request, transport booking, and handover summary.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Transfer form, Bed request, Transport book, Handover summary, Status track, Notify team, History, PDF export, Admin panel, Postman API

Problem 76: Medical Staff Credentialing Portal

Domain: Healthcare

Difficulty: ★★ Intermediate

Description:

Manage clinician licences, certifications, malpractice records, and background checks with expiry alerts and approval workflow.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Profile form, Document upload, Expiry alerts, Approval flow, Status tracker, History, Export PDF, Notifications, Admin panel, Postman API

Problem 77: Healthcare Analytics Dashboard

Domain: Healthcare

Difficulty: ★★★ Advanced

Description:

Executive dashboard for hospital KPIs — patient volumes, ALOS, readmission rates, revenue per bed, and departmental performance.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

KPI cards, Volume chart, ALOS trend, Readmission rate, Revenue chart, Dept compare, Date filter, Export, Alert config, Admin panel

Problem 78: Infection Control Tracker

Domain: Healthcare

Difficulty: ★ Beginner

Description:

Monitor and report healthcare-associated infections by ward with case counts, causative organisms, and outbreak detection alerts.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Case log form, Ward map, Organism chart, Outbreak alert, Case detail, Contact trace mock, History, Export, Notifications, Admin panel

Problem 79: Rehabilitation Progress Tracker

Domain: Healthcare

Difficulty: ★★ Intermediate

Description:

Track physiotherapy and rehabilitation milestones, exercise compliance, pain scores, and functional outcome measures.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Goal setting, Exercise log, Pain score, Milestone chart, Outcome measure, Therapist notes, History, Export PDF, Notifications, Admin panel

Problem 80: Ambulance Dispatch & Tracking UI

Domain: Healthcare

Difficulty: ★ Beginner

Description:

Dispatch ambulances for emergency calls, track real-time GPS location mock, assign paramedic crews, and log incident details.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Dispatch form, Map view, Crew assign, Status tracker, Incident log, Priority badge, History, Export, Notifications, Admin panel

Problem 81: Blood Bank Management System

Domain: Healthcare

Difficulty: ★★ Intermediate

Description:

Manage blood inventory by group and component, donor records, compatibility matching, issue requests, and expiry tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Inventory table, Donor profile, Compatibility check, Issue request, Expiry alert, Donation history, Export, Notifications, Admin panel, Postman API

Problem 82: Hospital Complaint & Feedback System

Domain: Healthcare

Difficulty: ★ ★ ★ Advanced

Description:

Patient and visitor complaint submission, tracking, escalation, and resolution with satisfaction scoring and trend reports.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Complaint form, Category select, Status track, Escalation flow, Resolution note, Satisfaction score, Analytics, Export, Notifications, Admin panel

Problem 83: Surgical Planning & Scheduling

Domain: Healthcare

Difficulty: ★ ★ Intermediate

Description:

Plan and schedule surgical procedures with surgeon availability, OT allocation, equipment checklist, and pre-op preparation tasks.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Procedure form, Surgeon calendar, OT assign, Equipment check, Pre-op list, Status tracker, Notify team, History, Export, Admin panel

Problem 84: Doctor Performance Dashboard

Domain: Healthcare

Difficulty: ★ ★ ★ Advanced

Description:

Track doctor performance metrics — patient load, satisfaction scores, consultation time, no-show rate, and billing productivity.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Metric cards, Trend charts, Patient list, Satisfaction score, Billing summary, Compare doctors, Date filter, Export, Alert config, Admin panel

Problem 85: Medical Waste Management Tracker

Domain: Healthcare

Difficulty: ★ Beginner

Description:

Log and track medical waste (sharps, biohazard, pharmaceutical) generation, collection, transport, and disposal compliance.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Waste log form, Category select, Transport assign, Disposal confirm, Compliance check, Chart analytics, History, Export, Notifications, Admin panel

Problem 86: Home Healthcare Visit Scheduler

Domain: Healthcare

Difficulty: ★ ★ Intermediate

Description:

Schedule and track home healthcare visits by nurses/therapists with patient location, task checklist, and visit notes.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Patient list, Visit scheduler, Task checklist, Map mock, Visit notes, Status update, History, Export, Notifications, Admin panel

Problem 87: Clinical Audit Management System

Domain: Healthcare

Difficulty: ★ ★ ★ Advanced

Description:

Plan and conduct clinical audits with data collection, benchmark comparison, action plans, and re-audit scheduling.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Audit form, Data entry, Benchmark display, Gap analysis, Action plan, Re-audit schedule, History, Export PDF, Notifications, Admin panel

Problem 88: Paediatric Growth Chart Monitor

Domain: Healthcare

Difficulty: ★ Beginner

Description:

Plot and track child growth measurements (weight, height, head circumference) against WHO percentile charts with alerts.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Measurement form, Percentile chart, Alert thresholds, Trend line, Comparison view, Export PDF, History, Profile, Notifications, Admin panel

Problem 89: Outpatient Waiting Time Monitor

Domain: Healthcare

Difficulty: ★★ Intermediate

Description:

Track and display outpatient waiting times by clinic and doctor with real-time queue number display and patient SMS mock.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Queue display, Clinic filter, Wait time chart, Queue number, SMS mock, Priority update, History, Export, Notifications, Admin panel

Problem 90: Healthcare Staff Training Portal

Domain: Healthcare

Difficulty: ★ Beginner

Description:

Assign, track, and certify mandatory and optional training modules for healthcare staff with quiz and completion badges.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Training catalogue, Module assign, Video/quiz UI, Score display, Certificate badge, Progress tracker, History, Notifications, Admin panel, Export

Problem 91: Medical Device Calibration Tracker

Domain: Healthcare

Difficulty: ★★ Intermediate

Description:

Schedule and record calibration of medical devices with due date alerts, technician assignments, and compliance certificates.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Device list, Calibration form, Due alerts, Technician assign, Certificate upload, Compliance status, History, Export, Notifications, Admin panel

Problem 92: Hospital Social Worker Case Manager

Domain: Healthcare

Difficulty: ★ ★ ★ Advanced

Description:

Manage social work cases — financial assistance, housing, discharge planning — with case notes, referrals, and outcome tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Case list, Notes form, Referral send, Outcome tracker, Document upload, Status update, History, Export, Notifications, Admin panel

Problem 93: Formulary Management System

Domain: Healthcare

Difficulty: ★ ★ Intermediate

Description:

Manage approved hospital drug formulary with addition/removal requests, therapeutic equivalents, cost comparison, and approval workflow.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Formulary list, Add request form, Therapeutic compare, Cost table, Approval workflow, Version history, Export, Notifications, Admin panel, Postman API

Problem 94: Organ Donation Registry UI

Domain: Healthcare

Difficulty: ★ ★ ★ Advanced

Description:

Register organ donors, manage consent, match potential donors to recipients, and track transplant coordination workflow.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Registration form, Consent upload, Match display, Status tracker, Notify team, History, FAQ, Export PDF, Admin panel, Postman API

Problem 95: Health Screening Camp Manager

Domain: Healthcare

Difficulty: ★ Beginner

Description:

Organise community health screening camps with participant registration, screening checklists, result recording, and follow-up referrals.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Camp form, Participant reg, Screening list, Result input, Referral generate, Summary report, Export, Notifications, Admin panel, Map view

Problem 96: Pathology Sample Tracking System

Domain: Healthcare

Difficulty: ★ ★ Intermediate

Description:

Track pathology samples from collection to result with chain of custody, rejection log, and turnaround time monitoring.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Sample log, Status timeline, Rejection form, TAT chart, Collector assign, Result link, History, Export, Notifications, Admin panel

Problem 97: Geriatric Care Assessment Tool

Domain: Healthcare

Difficulty: ★ ★ ★ Advanced

Description:

Comprehensive geriatric assessment tool for fall risk, cognitive screening, functional independence, and care plan generation.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Assessment form, Risk score, Cognitive screen, ADL checklist, Care plan, Alerts, History, Export PDF, Notifications, Admin panel

Problem 98: Medical Research Data Collection UI

Domain: Healthcare

Difficulty: ★ Beginner

Description:

Structured data collection forms for clinical research with branching logic, validation, audit trail, and export to CSV/Excel.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Form builder, Branch logic, Validation rules, Audit trail, Data entry UI, Progress tracker, Export CSV, Admin panel, Postman API, History

Problem 99: Hospital IT Helpdesk Portal

Domain: Healthcare

Difficulty: ★ ★ Intermediate

Description:

Internal IT helpdesk for hospital staff to log, track, and resolve hardware/software incidents and service requests.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Ticket form, Category select, Priority assign, Status track, Technician assign, Resolution note, SLA timer, History, Export, Admin panel

Education Domain

This section contains 50 complete hackathon problem statements for the Education Domain. Each problem is ready for immediate prototype development.

Problem 100: Student Admission Portal

Domain: Education

Difficulty: ★ Beginner

Description:

End-to-end admission application with eligibility check, document upload, fee payment mock, and admission status tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Application form, Eligibility check, Doc upload, Fee payment mock, Status tracker, Offer letter PDF, Profile, Notifications, Admin panel, Postman API

Problem 101: Online Learning Management System

Domain: Education

Difficulty: ★ ★ Intermediate

Description:

Course creation, student enrolment, content delivery (video/PDF), progress tracking, quizzes, and grading.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Course catalogue, Enrol flow, Content viewer, Progress bar, Quiz engine, Grade book, Certificate, Notifications, Admin panel, Postman API

Problem 102: Student Attendance Tracker

Domain: Education

Difficulty: ★ ★ ★ Advanced

Description:

Mark and track daily attendance by subject, generate reports, flag low-attendance students, and notify parents/guardians.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Attendance form, Class select, Mark UI, Report chart, Low-attendance alert, Parent notify mock, History, Export CSV, Admin panel, Profile

Problem 103: Online Examination Platform

Domain: Education

Difficulty: ★ ★ Intermediate

Description:

Timed online exams with MCQ/essay questions, auto-grading, anti-cheat indicators, result publication, and performance analytics.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Exam builder, Timer, MCQ engine, Essay input, Auto-grade, Anti-cheat flag, Result display, Analytics, PDF report, Admin panel

Problem 104: Assignment Submission System

Domain: Education

Difficulty: ★ ★ ★ Advanced

Description:

Submit and manage assignments with deadline management, plagiarism indicator mock, peer review, grading rubric, and feedback.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Submission portal, Deadline alert, File upload, Plagiarism mock, Peer review, Rubric grade, Feedback form, History, Export, Admin panel

Problem 105: Library Management System

Domain: Education

Difficulty: ★ Beginner

Description:

Search, reserve, borrow, and return books with overdue tracking, fine calculation, e-book access, and catalogue management.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Book search, Reservation form, Borrow/return log, Overdue alert, Fine calculator, E-book viewer mock, Catalogue admin, Export, Notifications, Postman API

Problem 106: Parent-Teacher Communication Portal

Domain: Education

Difficulty: ★★ Intermediate

Description:

Secure messaging between parents and teachers, meeting scheduler, report card sharing, and progress update notifications.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Message thread, Meeting scheduler, Report share, Notification, Read receipt, Announcement, Profile, Calendar view, Export, Admin panel

Problem 107: Timetable & Schedule Builder

Domain: Education

Difficulty: ★★★ Advanced

Description:

Create and publish class timetables with conflict detection, room assignment, teacher availability, and student view.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Timetable grid, Conflict alert, Room assign, Teacher calendar, Publish flow, Student view, Export PDF, Notification, History, Admin panel

Problem 108: Student Performance Analytics

Domain: Education

Difficulty: ★ Beginner

Description:

Visualise student academic performance over time with subject breakdowns, predicted grades, peer comparison, and intervention flags.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Performance dashboard, Subject chart, Trend line, Peer compare, Prediction display, Intervention flag, Export, Profile, Admin panel, Postman API

Problem 109: School Fee Management System

Domain: Education

Difficulty: ★★ Intermediate

Description:

Manage fee structures, generate invoices, record payments, send reminders for dues, and produce financial reports.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Fee structure config, Invoice generate, Payment record, Due reminder, Receipt PDF, Financial report, Parent portal, History, Export, Admin panel

Problem 110: Virtual Classroom UI

Domain: Education

Difficulty: ★ Beginner

Description:

Simulated online classroom with live session scheduling, whiteboard mock, breakout rooms, chat, polls, and recording link.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Session scheduler, Whiteboard mock, Chat panel, Poll widget, Breakout rooms, Recording link, Attendance mark, History, Notifications, Admin panel

Problem 111: Career Guidance & Counselling Portal

Domain: Education

Difficulty: ★ ★ Intermediate

Description:

Psychometric assessment tools, career path explorer, counsellor booking, resource library, and personalised roadmap.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Assessment form, Career explorer, Counsellor booking, Resource library, Roadmap display, Profile, History, Notifications, Export, Admin panel

Problem 112: Scholarship Management System

Domain: Education

Difficulty: ★ ★ ★ Advanced

Description:

Apply for scholarships, manage eligibility criteria, document submission, committee review, and disbursement tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Application form, Eligibility check, Doc upload, Review workflow, Decision notify, Disbursement track, History, Export PDF, Admin panel, Postman API

Problem 113: E-Portfolio Builder

Domain: Education

Difficulty: ★ ★ Intermediate

Description:

Students create digital portfolios with project showcases, skills, certifications, achievements, and shareable public link.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Portfolio builder, Project add, Skills section, Cert upload, Achievement badges, Public link, Privacy toggle, Profile, Export PDF, Admin panel

Problem 114: Peer Tutoring Marketplace

Domain: Education

Difficulty: ★ ★ ★ Advanced

Description:

Match students needing help with peer tutors by subject and level, schedule sessions, rate tutors, and track hours.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Tutor search, Subject filter, Session scheduler, Booking confirm, Rating form, Hour tracker, Payment mock, History, Notifications, Admin panel

Problem 115: School Event Management System

Domain: Education

Difficulty: ★ Beginner

Description:

Organise school events — sports day, cultural fest, parent days — with registration, schedule, volunteer management, and feedback.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Event list, Registration form, Schedule builder, Volunteer assign, Feedback form, Photo gallery mock, Notifications, History, Export, Admin panel

Problem 116: Digital Report Card Generator

Domain: Education

Difficulty: ★★ Intermediate

Description:

Generate and publish digital report cards with subject grades, teacher comments, attendance summary, and parent e-signature.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Grade input form, Comment editor, Attendance link, Template renderer, PDF generate, Parent notify, E-sign mock, History, Archive, Admin panel

Problem 117: Special Needs Education Tracker

Domain: Education

Difficulty: ★★★ Advanced

Description:

Track IEP goals, therapy sessions, accommodations, and progress for students with special educational needs.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

IEP form, Goal tracker, Therapy log, Accommodation notes, Progress chart, Parent share, History, Export PDF, Notifications, Admin panel

Problem 118: Campus Safety & Incident Reporter

Domain: Education

Difficulty: ★ Beginner

Description:

Report and track campus safety incidents — bullying, accidents, security — with investigation workflow and resolution tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Incident form, Category select, Investigation assign, Status track, Anonymous option, Resolution note, History, Export, Notifications, Admin panel

Problem 119: Student Feedback & Course Rating

Domain: Education

Difficulty: ★★ Intermediate

Description:

End-of-term course evaluations with quantitative ratings and qualitative feedback, results visible to faculty post-deadline.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Evaluation form, Rating scale, Comments, Anonymity toggle, Submission confirm, Results display, Faculty view, Analytics, Export, Admin panel

Problem 120: Academic Research Repository

Domain: Education

Difficulty: ★ Beginner

Description:

Upload, categorise, and search academic papers, dissertations, and research reports with citation generator and download tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Upload form, Category tags, Search, Citation generator, Download tracker, Preview, Profile, Notifications, Admin panel, Postman API

Problem 121: Hostel & Dormitory Management

Domain: Education

Difficulty: ★★ Intermediate

Description:

Allocate hostel rooms to students, manage check-in/out, maintenance requests, visitor logs, and fee collection.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Room allocation, Check-in form, Maintenance request, Visitor log, Fee record, Room map, History, Export, Notifications, Admin panel

Problem 122: Plagiarism Check UI Mock

Domain: Education

Difficulty: ★★★ Advanced

Description:

Interface for submitting documents for plagiarism analysis with similarity score display, highlighted matches, and source list.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Upload form, Processing animation, Score display, Highlight viewer, Source list, Report PDF, History, Notifications, Admin panel, Postman API

Problem 123: Student ID Card Generator

Domain: Education

Difficulty: ★★ Intermediate

Description:

Generate digital and printable student ID cards with photo, department, roll number, barcode, and validity period.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Student data form, Photo upload, Template designer, Barcode generate, PDF export, Batch generate, Preview, History, Admin panel, Notifications

Problem 124: Teacher Professional Development Portal

Domain: Education

Difficulty: ★★★ Advanced

Description:

Plan and track CPD activities, certifications, workshop attendance, reflection logs, and annual PD goal setting.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

PD catalogue, Activity log, Certificate upload, Goal setting, Reflection form, Progress chart, History, Export PDF, Notifications, Admin panel

Problem 125: Student Loan Application Portal

Domain: Education

Difficulty: ★ Beginner

Description:

Apply for educational loans with eligibility calculator, document upload, bank integration mock, status tracking, and repayment planner.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Eligibility calc, Application form, Doc upload, Bank mock, Status tracker, Repayment planner, Notifications, History, Export PDF, Admin panel

Problem 126: Class Participation Tracker

Domain: Education

Difficulty: ★★ Intermediate

Description:

Track and rate student participation in discussions, presentations, and collaborative work with teacher notes and trend charts.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Student list, Participation form, Rating scale, Notes field, Trend chart, Export report, Profile, History, Notifications, Admin panel

Problem 127: Curriculum Mapping Tool

Domain: Education

Difficulty: ★★★ Advanced

Description:

Map learning outcomes to curriculum units, assessment methods, and graduate attributes to ensure alignment and coverage.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Outcome list, Unit builder, Mapping grid, Coverage checker, Gap alert, Export PDF, History, Share link, Admin panel, Notifications

Problem 128: School Transport Route Manager

Domain: Education

Difficulty: ★ Beginner

Description:

Manage school bus routes, stops, student assignments, driver info, real-time mock tracking, and parent notifications.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Route builder, Stop list, Student assign, Driver profile, Map view, Parent notify, Delay alert, History, Export, Admin panel

Problem 129: Internship & Placement Portal

Domain: Education

Difficulty: ★★ Intermediate

Description:

Post and apply for internships/jobs, manage placement drives, interview scheduling, offer tracking, and placement statistics.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Job listing, Application form, Drive schedule, Interview slot, Offer tracker, Placement stats, Profile, Notifications, Export, Admin panel

Problem 130: Gamified Learning Progress Tracker

Domain: Education

Difficulty: ★ Beginner

Description:

Earn XP, badges, and level up based on course completions, quiz scores, and streaks in a gamified academic interface.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

XP tracker, Badge gallery, Level display, Leaderboard, Streak counter, Course link, Profile, Notifications, Export, Admin panel

Problem 131: Alumni Network Portal

Domain: Education

Difficulty: ★★ Intermediate

Description:

Connect alumni with current students for mentorship, job referrals, events, donations, and batch reunion management.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Profile directory, Mentor connect, Job post, Event list, Donation mock, Batch groups, Messaging, Notifications, Export, Admin panel

Problem 132: School Newsletter Creator

Domain: Education

Difficulty: ★ ★ ★ Advanced

Description:

Design, publish, and distribute school newsletters with a drag-and-drop builder, article submission, and email distribution.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Article submission, Drag-drop builder, Preview, Publish flow, Email distribute mock, Archive, Category filter, Profile, Admin panel, Notifications

Problem 133: Student Mental Health Support Hub

Domain: Education

Difficulty: ★ ★ Intermediate

Description:

Anonymous mental health support portal with mood check-ins, journaling, resource articles, counsellor requests, and crisis links.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Mood check-in, Journal form, Resource library, Counsellor request, Crisis links, Anonymity toggle, History, Notifications, Profile, Admin panel

Problem 134: Academic Calendar Builder

Domain: Education

Difficulty: ★ ★ ★ Advanced

Description:

Create and publish institution-wide academic calendars with events, holidays, exam dates, and department-specific overlays.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Calendar grid, Event add form, Holiday flag, Exam dates, Dept overlay, Publish flow, Export PDF, Subscribe mock, Notifications, Admin panel

Problem 135: Reading Progress & Book Log

Domain: Education

Difficulty: ★ Beginner

Description:

Log books read, rate them, write reviews, track reading goals, and view reading history with trend charts and recommendations.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Book log form, Rating input, Review editor, Goal tracker, Trend chart, Recommendation mock, Profile, History, Export, Admin panel

Problem 136: Teacher Lesson Plan Builder

Domain: Education

Difficulty: ★ ★ Intermediate

Description:

Create, store, and share lesson plans with objectives, activities, resources, assessment methods, and curriculum alignment.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Plan form, Objective builder, Activity list, Resource attach, Assessment link, Curriculum tag, Share link, History, Export PDF, Admin panel

Problem 137: Student Behaviour Incident Log

Domain: Education

Difficulty: ★ ★ ★ Advanced

Description:

Record and manage student behaviour incidents with severity rating, parent notification, counsellor referral, and follow-up tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Incident form, Severity badge, Parent notify mock, Counsellor referral, Follow-up tracker, History, Export, Notifications, Admin panel, Profile

Problem 138: Online STEM Lab Simulator UI

Domain: Education

Difficulty: ★ Beginner

Description:

Simulated experiments for physics/chemistry labs with parameter inputs, result calculation, and lab report generation.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Experiment selector, Parameter form, Simulation canvas, Result display, Lab report generator, Theory section, History, Export PDF, Admin panel, Notifications

Problem 139: Student Wellness & Fitness Tracker

Domain: Education

Difficulty: ★★ Intermediate

Description:

Track physical activity, sleep, hydration, and wellness scores with weekly goals, progress charts, and peer challenges.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Activity log, Sleep tracker, Hydration input, Wellness score, Goal setter, Progress chart, Peer challenge, Notifications, History, Admin panel

Problem 140: School Procurement Portal

Domain: Education

Difficulty: ★ Beginner

Description:

Raise and approve purchase requests for school supplies, labs, and IT equipment with vendor comparison and budget control.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

PO form, Approval flow, Vendor compare, Budget tracker, Delivery confirm, Invoice upload, History, Export, Notifications, Admin panel

Problem 141: Digital Classroom Whiteboard

Domain: Education

Difficulty: ★★ Intermediate

Description:

Collaborative whiteboard for live teaching with shapes, text, sticky notes, pointer, undo/redo, and session save.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Canvas UI, Shape tools, Text tool, Sticky notes, Pointer, Undo/redo, Save session, Export image, Multi-user mock, Admin panel

Problem 142: Language Learning Flashcard App

Domain: Education

Difficulty: ★ ★ ★ Advanced

Description:

Spaced repetition flashcard system for vocabulary learning with audio mock, image association, quiz mode, and progress tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Card creator, SRS engine, Quiz mode, Audio mock, Image attach, Progress chart, Deck manage, History, Notifications, Export

Problem 143: School Canteen Pre-Order System

Domain: Education

Difficulty: ★ ★ Intermediate

Description:

Pre-order canteen meals by day with menu display, dietary filters, payment mock, collection QR code, and feedback.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Menu display, Dietary filter, Cart, Payment mock, Collection QR, Order history, Feedback form, Notifications, Admin panel, Export

Problem 144: Teacher Evaluation Portal

Domain: Education

Difficulty: ★ ★ ★ Advanced

Description:

360-degree teacher evaluation by students, peers, and administrators with rubric scoring, trend analysis, and development plans.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Evaluation form, Rubric builder, Score display, Trend chart, Peer review, Admin review, Development plan, History, Export, Notifications

Problem 145: School Management Admin Panel

Domain: Education

Difficulty: ★ Beginner

Description:

Comprehensive admin dashboard for school management — enrolment, staff, finance, events, and performance overview.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Dashboard KPIs, Enrolment chart, Staff list, Finance summary, Event calendar, Performance metrics, Export, Alert config, Notifications, Postman API

Problem 146: STEM Competition Registration

Domain: Education

Difficulty: ★ ★ Intermediate

Description:

Register teams for science/tech competitions with member details, project abstract, mentor assign, submission, and result declaration.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Team form, Member add, Abstract upload, Mentor assign, Submission portal, Result display, Certificate mock, History, Notifications, Admin panel

Problem 147: Student Disciplinary Records System

Domain: Education

Difficulty: ★ ★ ★ Advanced

Description:

Maintain confidential disciplinary records with incident logs, hearing notes, sanctions, appeal submissions, and closure records.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Case log, Hearing notes, Sanction record, Appeal form, Status tracker, Confidentiality flag, History, Export PDF, Notifications, Admin panel

Problem 148: School Website CMS

Domain: Education

Difficulty: ★ Beginner

Description:

Content management system for school website with page builder, news section, staff directory, event calendar, and image gallery.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Page builder, News editor, Staff directory, Event calendar, Gallery upload, Publish flow, SEO mock, History, Notifications, Admin panel

Problem 149: Distance Learning Analytics Dashboard

Domain: Education

Difficulty: ★ ★ Intermediate

Description:

Analyse online learning engagement — video completion, quiz attempts, login frequency, and learning time — with intervention alerts.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Engagement chart, Completion rate, Quiz attempts, Login heatmap, Learning time, Intervention flag, Export, Date filter, Admin panel, Postman API



Hotels Domain

This section contains 50 complete hackathon problem statements for the Hotels Domain. Each problem is ready for immediate prototype development.

Problem 150: Hotel Room Booking Engine

Domain: Hotels

Difficulty: ★ Beginner

Description:

Search and book hotel rooms by date, occupancy, and amenities with real-time availability, pricing, and instant confirmation.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Room search, Date picker, Occupancy select, Filter panel, Room detail, Booking form, Payment mock, Confirmation, History, Admin panel

Problem 151: Hotel Check-In & Check-Out Portal

Domain: Hotels

Difficulty: ★ ★ Intermediate

Description:

Self-service online check-in with ID upload, room assignment, digital key mock, and express check-out with invoice generation.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Check-in form, ID upload, Room assign, Digital key mock, Check-out flow, Invoice PDF, Guest profile, Notifications, History, Admin panel

Problem 152: Hotel Room Service Order System

Domain: Hotels

Difficulty: ★ ★ ★ Advanced

Description:

In-room dining and amenity request portal with menu browsing, order placement, delivery tracking, and billing.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Menu display, Cart, Order form, Delivery tracker, Billing summary, Order history, Rating form, Notifications, Admin panel, Postman API

Problem 153: Event & Banquet Booking System

Domain: Hotels

Difficulty: ★★ Intermediate

Description:

Book hotel banquet halls and meeting rooms with capacity configuration, AV equipment selection, catering packages, and décor options.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Venue catalogue, Capacity filter, AV selector, Catering packages, Décor options, Booking form, Payment mock, Confirmation, History, Admin panel

Problem 154: Guest Loyalty Program Dashboard

Domain: Hotels

Difficulty: ★★★ Advanced

Description:

Manage hotel loyalty points, tier status, benefits, redemption for free nights/upgrades, and partner rewards.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Points dashboard, Tier display, Benefits list, Redeem flow, Partner rewards, History, Profile, Notifications, Export, Admin panel

Problem 155: Hotel Housekeeping Management

Domain: Hotels

Difficulty: ★ Beginner

Description:

Assign and track room cleaning tasks by floor/room status (dirty, clean, occupied, vacant) with housekeeper mobile view.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Room status grid, Task assign, Status update, Floor map, Priority flag, Completion log, Housekeeper view, History, Export, Admin panel

Problem 156: Hotel Revenue Management Dashboard

Domain: Hotels

Difficulty: ★★ Intermediate

Description:

Track ADR, RevPAR, occupancy rate, and booking pace with forecast vs actuals, channel breakdown, and pricing recommendations.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

KPI cards, ADR chart, RevPAR trend, Occupancy gauge, Forecast display, Channel breakdown, Pricing suggest, Export, Date filter, Admin panel

Problem 157: Guest Feedback & Review Portal

Domain: Hotels

Difficulty: ★★★ Advanced

Description:

Collect guest satisfaction surveys post-stay with NPS scoring, category ratings, comment moderation, and response workflow.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Survey form, NPS score, Category rating, Comment input, Moderation queue, Response template, Analytics chart, Export, Notifications, Admin panel

Problem 158: Hotel Restaurant Reservation System

Domain: Hotels

Difficulty: ★ Beginner

Description:

Reserve tables at hotel restaurants with party size, time slots, dietary preferences, special occasion notes, and waitlist management.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Table map, Time slot, Party size, Dietary note, Special occasion, Waitlist, Confirm email mock, History, Admin panel, Postman API

Problem 159: Concierge Service Request Portal

Domain: Hotels

Difficulty: ★★ Intermediate

Description:

Submit and track concierge requests — taxi booking, tour tickets, recommendations — with fulfilment status and guest messaging.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Request form, Category select, Status tracker, Message thread, Fulfilment confirm, History, Profile, Notifications, Admin panel, Export

Problem 160: Hotel Channel Manager UI

Domain: Hotels

Difficulty: ★ Beginner

Description:

Manage room inventory and pricing across OTA channels (mock Booking.com, Expedia) with rate parity alerts and bulk updates.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Channel list, Inventory grid, Rate input, Parity alert, Bulk update, Sync status, History, Export, Notifications, Admin panel

Problem 161: Spa & Wellness Booking System

Domain: Hotels

Difficulty: ★ ★ Intermediate

Description:

Book spa treatments, fitness classes, and wellness sessions with therapist selection, time slots, package deals, and gift vouchers.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Treatment catalogue, Therapist select, Time slot, Package picker, Gift voucher, Payment mock, Confirmation, History, Notifications, Admin panel

Problem 162: Hotel Maintenance & Repair Tracker

Domain: Hotels

Difficulty: ★ ★ ★ Advanced

Description:

Log and track maintenance requests from staff or guests with technician assignment, priority, resolution, and cost tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Request form, Priority badge, Technician assign, Status update, Cost input, Resolution note, History, Export, Notifications, Admin panel

Problem 163: Meeting & Conference Room Scheduler

Domain: Hotels

Difficulty: ★ ★ Intermediate

Description:

Book meeting rooms with capacity, AV requirements, catering add-ons, and real-time availability calendar for corporate clients.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Room list, Capacity filter, AV select, Catering add-on, Calendar view, Booking form, Confirm, History, Notifications, Admin panel

Problem 164: Hotel Procurement & Vendor Portal

Domain: Hotels

Difficulty: ★ ★ ★ Advanced

Description:

Manage purchasing of hotel supplies from vendors with RFQ, PO creation, delivery tracking, invoice processing, and vendor ratings.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Vendor list, RFQ form, PO creator, Delivery track, Invoice upload, Rating form, History, Export, Notifications, Admin panel

Problem 165: VIP Guest Management System

Domain: Hotels

Difficulty: ★ Beginner

Description:

Manage VIP and repeat guest profiles with preference tracking, amenity pre-setting, arrival alerts, and personalised welcome messages.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Guest CRM, Preference profile, Amenity pre-set, Arrival alert, Welcome message, Booking link, History, Export, Notifications, Admin panel

Problem 166: Hotel Digital Signage Content Manager

Domain: Hotels

Difficulty: ★★ Intermediate

Description:

Create and schedule digital signage content for lobby screens — welcome messages, event listings, weather, promotions, and menus.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Content builder, Schedule manager, Screen assign, Preview, Publish flow, Weather widget mock, History, Notifications, Admin panel, Export

Problem 167: Hotel Staff Onboarding Portal

Domain: Hotels

Difficulty: ★★★ Advanced

Description:

Onboard new hotel staff with department-specific training modules, uniform requests, system access provisioning, and HR task checklist.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Onboard form, Training list, Uniform request, Access provision, HR checklist, Progress tracker, Profile, Notifications, Export, Admin panel

Problem 168: Hotel Complaint Escalation System

Domain: Hotels

Difficulty: ★ Beginner

Description:

Track guest complaints from report to resolution with escalation to department heads, compensation offers, and satisfaction follow-up.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Complaint form, Category, Escalation flow, Assign head, Comp offer, Resolution note, Satisfaction score, History, Export, Admin panel

Problem 169: Hotel Group Booking Manager

Domain: Hotels

Difficulty: ★★ Intermediate

Description:

Handle group reservations (MICE, weddings, tours) with rooming lists, billing splits, contract generation, and room block management.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Group form, Rooming list, Block manage, Billing split, Contract PDF, Status track, Email all, History, Export, Admin panel

Problem 170: Hotel Energy Management Dashboard

Domain: Hotels

Difficulty: ★ Beginner

Description:

Monitor energy consumption (electricity, water, gas) by department with usage trends, benchmarks, and sustainability goals.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Usage dashboard, Trend chart, Dept breakdown, Benchmark display, Goal tracker, Alert config, Export, History, Notifications, Admin panel

Problem 171: Hotel Gift Shop & Retail POS UI

Domain: Hotels

Difficulty: ★★ Intermediate

Description:

Point-of-sale interface for hotel gift shop with product catalogue, barcode lookup, cart, payment, receipt, and inventory update.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Product catalogue, Barcode lookup, Cart, Payment form, Receipt print mock, Inventory update, History, Export, Notifications, Admin panel

Problem 172: Pool & Recreation Booking System

Domain: Hotels

Difficulty: ★★★ Advanced

Description:

Reserve pool cabanas, tennis courts, golf tee times, and activities with time slots, guest count, and add-on equipment.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Activity catalogue, Slot picker, Guest count, Equipment add-on, Payment mock, Confirmation, History, Notifications, Admin panel, Export

Problem 173: Hotel Night Audit Dashboard

Domain: Hotels

Difficulty: ★ ★ Intermediate

Description:

End-of-day reconciliation of room revenues, F&B, and payments with variance analysis, balance report, and journal entries mock.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Revenue summary, F&B breakdown, Payment reconcile, Variance chart, Balance report, Journal mock, Export PDF, History, Notifications, Admin panel

Problem 174: Guest Room Customisation Portal

Domain: Hotels

Difficulty: ★ ★ ★ Advanced

Description:

Pre-arrival room customisation — pillow preference, room temperature, music, mini-bar preferences — synced to property management.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Preference form, Category select, Sync mock, Confirmation, Profile, History, Notifications, Admin panel, Edit flow, Export

Problem 175: Hotel Job Recruitment Portal

Domain: Hotels

Difficulty: ★ Beginner

Description:

Post hotel positions, collect applications, schedule interviews, manage candidate pipeline, and generate offer letters.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Job posting, Application form, Pipeline view, Interview scheduler, Offer letter PDF, Status track, Profile, Notifications, Export, Admin panel

Problem 176: Hotel F&B Cost Control Dashboard

Domain: Hotels

Difficulty: ★★ Intermediate

Description:

Track food and beverage costs, recipe costing, waste recording, menu profitability, and supplier price monitoring.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Cost dashboard, Recipe costing, Waste log, Menu profitability chart, Supplier price, Variance alert, Export, History, Notifications, Admin panel

Problem 177: Hotel Shuttle & Transport Scheduler

Domain: Hotels

Difficulty: ★★★ Advanced

Description:

Schedule and track hotel shuttle services to/from airport and attractions with guest booking, driver assign, and live status mock.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Shuttle schedule, Guest booking, Driver assign, Status mock, Route display, Confirm notify, History, Export, Notifications, Admin panel

Problem 178: Hotel Website CMS & Booking Widget

Domain: Hotels

Difficulty: ★ Beginner

Description:

Manage hotel website content with room descriptions, photo gallery, SEO metadata, offers, and embedded direct booking widget.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Page editor, Room listing, Gallery upload, Offer builder, SEO fields, Booking widget, Preview, Publish, History, Admin panel

Problem 179: Hotel Crisis & Emergency Management

Domain: Hotels

Difficulty: ★★ Intermediate

Description:

Emergency response dashboard for hotels — fire, medical, security incidents — with alert broadcasting, evacuation tracking, and contact tree.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Incident form, Alert broadcast, Contact tree, Evacuation tracker, Status board, Notify authorities mock, History, Export, Notifications, Admin panel

Problem 180: Guest Itinerary Builder

Domain: Hotels

Difficulty: ★ Beginner

Description:

Concierge-assisted or self-service tool for building personalised sightseeing itineraries with hotel tie-in activities and transport.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Itinerary builder, Activity search, Map view, Hotel activities, Transport add, Time planner, PDF export, Share link, History, Admin panel

Problem 181: Hotel Loyalty Partnerships Manager

Domain: Hotels

Difficulty: ★★ Intermediate

Description:

Manage earn-and-burn partnerships with airlines, car rentals, and retail brands with API mock, conversion rates, and reporting.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Partner list, Earn config, Burn config, Conversion calc, API mock, Report chart, History, Export, Notifications, Admin panel

Problem 182: Hotel Rate Shopper UI

Domain: Hotels

Difficulty: ★★ ★ Advanced

Description:

Compare hotel rates against competitors on OTAs with date range, room type comparison, parity alerts, and historical rate trends.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Rate table, Competitor list, OTA mock, Parity alert, Historical chart, Date range, Export, Notifications, Admin panel, Filter panel

Problem 183: Hotel Payroll & Scheduling System

Domain: Hotels

Difficulty: ★★ Intermediate

Description:

Manage hotel staff schedules, time and attendance, overtime, and payroll calculation with department-wise reporting.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Schedule builder, Timesheet, Overtime calc, Payroll summary, Dept report, Leave link, Export, Notifications, History, Admin panel

Problem 184: Hotel In-Room Technology Controller

Domain: Hotels

Difficulty: ★★ ★ Advanced

Description:

Smart room interface mockup controlling lights, blinds, TV, AC, and DND/MUR status from a tablet-style UI.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Room controls UI, Light dimmer, AC control, TV remote mock, Blinds control, DND toggle, Status display, History, Profile, Admin panel

Problem 185: Hotel Photo & Media Manager

Domain: Hotels

Difficulty: ★ Beginner

Description:

Upload, tag, and manage hotel photography for website, OTAs, and social media with resolution check and watermarking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Photo upload, Category tag, Resolution check, Watermark apply, Gallery view, OTA export, Social share mock, History, Admin panel, Notifications

Problem 186: Hotel Bar & Minibar Tracking

Domain: Hotels

Difficulty: ★★ Intermediate

Description:

Track minibar consumption, beverage inventory, replenishment schedules, breakage recording, and billing reconciliation.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Minibar log, Consumption form, Replenish schedule, Breakage record, Billing link, Inventory chart, History, Export, Notifications, Admin panel

Problem 187: Hotel Fire Safety Compliance Tracker

Domain: Hotels

Difficulty: ★★★ Advanced

Description:

Schedule and record fire safety drills, extinguisher checks, alarm tests, and staff certification with regulatory deadline alerts.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Compliance checklist, Drill log, Equipment check, Test record, Cert upload, Deadline alerts, History, Export PDF, Notifications, Admin panel

Problem 188: Hotel Social Media Management UI

Domain: Hotels

Difficulty: ★ Beginner

Description:

Schedule and publish social media posts for hotel channels with content calendar, engagement tracker, and response templates.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Post scheduler, Content calendar, Channel select, Engagement stats, Response templates, Media upload, History, Analytics, Notifications, Admin panel

Problem 189: Hotel Guest Arrival Intelligence

Domain: Hotels

Difficulty: ★ ★ Intermediate

Description:

Pre-arrival dashboard showing upcoming guest profiles, preferences, loyalty tier, special requests, and room assignment recommendations.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Arrival list, Guest CRM view, Preference display, Special request, Room recommend, Assign confirm, VIP flag, History, Export, Admin panel

Problem 190: Hotel Package & Offer Builder

Domain: Hotels

Difficulty: ★ Beginner

Description:

Create promotional packages (honeymoon, business, staycation) with room + dining + activity bundles, pricing, and validity management.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Package builder, Bundle select, Price input, Validity dates, Preview, Publish flow, Performance chart, History, Notifications, Admin panel

Problem 191: Hotel Accessibility Services Manager

Domain: Hotels

Difficulty: ★ ★ Intermediate

Description:

Manage accessibility requests — roll-in shower, visual/hearing impairment aids — with room allocation and staff briefing.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Request form, Aid type select, Room allocate, Staff brief, Status track, Guest confirm, History, Export, Notifications, Admin panel

Problem 192: Hotel Payment & Folio System

Domain: Hotels

Difficulty: ★★ ★ Advanced

Description:

Manage guest folios — room charges, dining, services — with split billing, corporate invoicing, deposit management, and checkout settle.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Folio view, Charge add, Split bill, Corporate invoice, Deposit tracker, Settle form, Receipt PDF, History, Export, Admin panel

Problem 193: Hotel Parking Management System

Domain: Hotels

Difficulty: ★★ Intermediate

Description:

Manage hotel parking bays with valet assignment, vehicle log, parking charges, space availability, and monthly permit management.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Parking map, Valet assign, Vehicle log, Charge calc, Availability display, Permit manage, History, Export, Notifications, Admin panel

Problem 194: Hotel Lost Property Register

Domain: Hotels

Difficulty: ★★ ★ Advanced

Description:

Log, match, and return lost items found on hotel premises with guest inquiry, storage location, and courier arrangement.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Lost log form, Found item list, Match search, Guest inquiry, Storage assign, Courier form, History, Export, Notifications, Admin panel

Problem 195: Hotel Tender & Contract Manager

Domain: Hotels

Difficulty: ★ Beginner

Description:

Manage procurement tenders for hotel supplies and services with vendor shortlisting, bid comparison, contract upload, and renewal alerts.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Tender form, Vendor list, Bid compare, Contract upload, Renewal alert, Status track, History, Export, Notifications, Admin panel

Problem 196: Hotel Performance KPI Tracker

Domain: Hotels

Difficulty: ★ ★ Intermediate

Description:

Track hotel performance KPIs — GSS score, RevPAR, EBITDA, staff turnover — with monthly targets, variance analysis, and board report.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

KPI dashboard, Target vs actual, Variance chart, Department breakdown, Monthly compare, Board report PDF, Notifications, History, Export, Admin panel

Problem 197: Hotel Laundry & Linen Management

Domain: Hotels

Difficulty: ★ ★ ★ Advanced

Description:

Track laundry requests, linen inventory by room type, turnover schedules, vendor laundry service, and damage recording.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Laundry request, Linen inventory, Turnover schedule, Vendor assign, Damage log, Status track, History, Export, Notifications, Admin panel

Problem 198: Hotel Mini-Conference Booking Portal

Domain: Hotels

Difficulty: ★ Beginner

Description:

Book small meeting pods, boardrooms, and breakout spaces with hourly pricing, whiteboard/AV options, and catering add-ons.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Space catalogue, Hourly pricing, AV option, Catering add-on, Booking form, Payment mock, Confirmation, History, Notifications, Admin panel



College Domain

This section contains 50 complete hackathon problem statements for the College Domain. Each problem is ready for immediate prototype development.

Problem 199: College Admission Management System

Domain: College

Difficulty: ★ Beginner

Description:

Full-cycle college admission portal with application forms, document upload, merit ranking, interview scheduling, and seat allotment.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Application form, Doc upload, Merit rank, Interview scheduler, Allotment display, Offer letter PDF, Status track, Notifications, Admin panel, Postman API

Problem 200: Student Academic Records System

Domain: College

Difficulty: ★ ★ Intermediate

Description:

Maintain semester-wise marks, grade calculation (GPA/CGPA), backlogs, re-valuation requests, and transcript generation.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Marks entry, Grade calc, CGPA display, Backlog flag, Re-valuation form, Transcript PDF, History, Profile, Notifications, Admin panel

Problem 201: Faculty Management Portal

Domain: College

Difficulty: ★ ★ ★ Advanced

Description:

Manage faculty profiles, subject assignments, workload tracking, research publications, leave requests, and performance reviews.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Faculty profile, Subject assign, Workload chart, Publication list, Leave form, Performance review, History, Export, Notifications, Admin panel

Problem 202: College ERP Dashboard

Domain: College

Difficulty: ★★ Intermediate

Description:

Integrated college ERP overview — enrolment, finance, academics, HR, and infrastructure — with role-based access.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Dashboard KPIs, Module links, Role-based view, Notification centre, Quick actions, Alert config, History, Export, Admin panel, Postman API

Problem 203: Student Grievance Redressal Portal

Domain: College

Difficulty: ★★★ Advanced

Description:

Submit academic, administrative, and personal grievances with category tagging, escalation workflow, and resolution timeline.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Grievance form, Category select, Escalation flow, Assign officer, Status track, Resolution note, Anonymous option, History, Export, Admin panel

Problem 204: Research Project Management Portal

Domain: College

Difficulty: ★ Beginner

Description:

Plan and track research projects with milestones, budget tracking, publication links, ethics approval, and collaboration management.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Project form, Milestone planner, Budget tracker, Publication link, Ethics upload, Collaborator add, Status track, Export PDF, Notifications, Admin panel

Problem 205: College Transport Management

Domain: College

Difficulty: ★★ Intermediate

Description:

Manage college bus routes, student bus pass applications, route changes, driver tracking mock, and payment collection.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Route builder, Bus pass form, Route change, Map mock, Driver profile, Payment record, History, Export, Notifications, Admin panel

Problem 206: HOD & Dean Approval Workflow

Domain: College

Difficulty: ★★★ Advanced

Description:

Multi-level document approval system for leave requests, project proposals, budget sanctions, and event permissions.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Request form, Approval chain, Status display, Comment thread, Notify approver, Escalation, History, Export, Notifications, Admin panel

Problem 207: College Canteen Token System

Domain: College

Difficulty: ★ Beginner

Description:

Pre-order canteen meals with token generation, menu display, dietary preferences, daily limits, and billing dashboard.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Menu display, Token form, Dietary filter, Daily limit, Payment mock, Token QR, Billing dashboard, History, Notifications, Admin panel

Problem 208: College Sports & Extracurricular Portal

Domain: College

Difficulty: ★★ Intermediate

Description:

Register and manage sports teams, cultural clubs, event participation, achievement records, and intramural scheduling.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Club list, Registration form, Team manage, Achievement log, Event schedule, Participation cert, History, Notifications, Admin panel, Export

Problem 209: Guest Lecture & Seminar Manager

Domain: College

Difficulty: ★ Beginner

Description:

Organise external guest lectures and seminars with speaker profiles, registration, attendance, certificates, and feedback collection.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Event form, Speaker profile, Registration, Attendance mark, Certificate PDF, Feedback form, History, Notifications, Export, Admin panel

Problem 210: Anti-Ragging Complaint Portal

Domain: College

Difficulty: ★ ★ Intermediate

Description:

Anonymous anti-ragging complaint submission with case tracking, committee assignment, and statutory report generation.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Complaint form, Anonymity option, Case assign, Status track, Committee notes, Report generate, History, Export, Notifications, Admin panel

Problem 211: College Alumni Placement Records

Domain: College

Difficulty: ★ ★ ★ Advanced

Description:

Track and display alumni career trajectories, company placements, salary bands, and higher education outcomes for prospective students.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Alumni list, Placement record, Salary chart, Company display, Higher ed track, Filter panel, Profile, History, Export, Admin panel

Problem 212: Internal Quality Assurance Cell (IQAC) Portal

Domain: College

Difficulty: ★★ Intermediate

Description:

Manage NAAC/NBA accreditation documentation, annual quality reports, criterion-wise data entry, and faculty development tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Criterion form, Doc upload, AQR generator, Faculty dev log, Criterion dashboard, Compliance check, Export PDF, History, Notifications, Admin panel

Problem 213: Departmental Budget Tracker

Domain: College

Difficulty: ★★★ Advanced

Description:

Department heads track budget allocations, expenditures, reimbursement requests, and end-of-year utilisation reports.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Budget summary, Expenditure log, Reimbursement form, Utilisation chart, Alert config, Export, History, Notifications, Admin panel, Approval flow

Problem 214: College ID Card & Certificate Issuance

Domain: College

Difficulty: ★ Beginner

Description:

Apply for bonafide certificates, NOC, experience letters, and replacement ID cards with approval and collection tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Certificate form, Type select, Approval flow, Status track, Collection notify, PDF generate, History, Admin panel, Notifications, Export

Problem 215: Examination Invigilation Duty Allocator

Domain: College

Difficulty: ★ ★ Intermediate

Description:

Assign and publish exam invigilation duties to faculty with conflict check, substitution requests, and duty acknowledgement.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Faculty list, Duty assign, Conflict check, Sub request, Acknowledgement form, Export schedule, History, Notifications, Admin panel, Calendar view

Problem 216: College News & Notice Board

Domain: College

Difficulty: ★ ★ ★ Advanced

Description:

Publish and manage college circulars, notices, news articles, and urgent alerts with category filter and role-based visibility.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Notice form, Category tag, Role filter, Alert flag, Rich text editor, Publish flow, Archive, Search, Notifications, Admin panel

Problem 217: Student Project Allocation System

Domain: College

Difficulty: ★ Beginner

Description:

Allocate final year projects to students with guide assignment, domain selection, preference matching, and progress milestones.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Project list, Preference form, Guide assign, Allocation display, Milestone tracker, Report upload, History, Notifications, Export, Admin panel

Problem 218: College Inventory & Asset Management

Domain: College

Difficulty: ★ ★ Intermediate

Description:

Track college assets — furniture, equipment, computers — with purchase records, condition updates, repair logs, and disposal workflow.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Asset list, Purchase record, Condition update, Repair log, Disposal form, Location map, History, Export, Notifications, Admin panel

Problem 219: NSS/NCC Activity Tracker

Domain: College

Difficulty: ★ Beginner

Description:

Log NSS/NCC activities, camp attendance, community service hours, certificates, and annual programme completion.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Activity log, Attendance form, Hour tracker, Certificate upload, Annual report, Profile, History, Export, Notifications, Admin panel

Problem 220: College Placement Training Portal

Domain: College

Difficulty: ★ ★ Intermediate

Description:

Manage pre-placement training sessions, aptitude resources, mock tests, company-specific prep, and training attendance.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Session list, Aptitude resources, Mock test, Company prep, Attendance mark, Score tracker, History, Export, Notifications, Admin panel

Problem 221: Faculty Leave & Duty Management

Domain: College

Difficulty: ★ ★ ★ Advanced

Description:

Faculty submit and track leave applications, OD requests, and compensatory duty with HOD approval and academic impact check.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Leave form, OD form, Comp duty, Approval chain, Academic impact check, Calendar view, History, Export, Notifications, Admin panel

Problem 222: College Annual Report Generator

Domain: College

Difficulty: ★ ★ Intermediate

Description:

Compile and generate annual institutional report from structured data inputs across departments with charts and executive summary.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Data input modules, Chart generator, Summary editor, Dept sections, PDF renderer, Version control, Export, History, Notifications, Admin panel

Problem 223: College Cultural Fest Registration

Domain: College

Difficulty: ★ ★ ★ Advanced

Description:

Register students and college teams for cultural fest events with team formation, event category selection, and stage scheduling.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Event catalogue, Team form, Member add, Stage schedule, QR pass, Score entry, Results display, History, Notifications, Admin panel

Problem 224: Smart Classroom Booking System

Domain: College

Difficulty: ★ Beginner

Description:

Reserve smart classrooms, labs, and AV-equipped rooms for classes, events, and extra sessions with conflict detection.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Room list, Capacity filter, AV check, Calendar view, Booking form, Conflict alert, Confirm notify, History, Export, Admin panel

Problem 225: College Health Centre Portal

Domain: College

Difficulty: ★★ Intermediate

Description:

Student health consultation booking, medical certificate issuance, pharmacy stock, illness leave link, and vaccination record.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Appointment book, Consultation notes, Med cert generate, Pharmacy stock, Illness leave link, Vaccination record, History, Notifications, Admin panel, Export

Problem 226: Research Ethics Committee Portal

Domain: College

Difficulty: ★★★ Advanced

Description:

Submit research proposals for ethical review with human/animal subject forms, committee assignment, and approval tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Proposal form, Ethics checklist, Reviewer assign, Status track, Decision notify, Revision submission, History, Export PDF, Notifications, Admin panel

Problem 227: College Accreditation Self-Study Report

Domain: College

Difficulty: ★ Beginner

Description:

Build the NAAC/NBA self-study report criterion by criterion with document attachments, metric entries, and completeness tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Criterion dashboard, Metric entry, Doc attach, Completeness bar, Review flow, Export PDF, Version history, Notifications, Admin panel, Postman API

Problem 228: Student Mentorship Programme Tracker

Domain: College

Difficulty: ★★ Intermediate

Description:

Assign mentors to students, log mentoring sessions, track academic and personal progress, and generate mentoring reports.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Mentor assign, Session log, Student progress, Issue flag, Report generate, Meeting scheduler, History, Export, Notifications, Admin panel

Problem 229: Department Website CMS

Domain: College

Difficulty: ★ Beginner

Description:

Content management for individual department websites — faculty profiles, research areas, events, courses, and news.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Page builder, Faculty profiles, Research area, Event list, Course display, News editor, Publish flow, Preview, History, Admin panel

Problem 230: Thesis & Dissertation Submission Portal

Domain: College

Difficulty: ★★ Intermediate

Description:

Submit and track PhD/PG dissertations with supervisor approval, plagiarism check mock, format validation, and library archival.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Submission form, Supervisor review, Plagiarism mock, Format check, Status track, Archive link, Certificate, History, Notifications, Admin panel

Problem 231: College Feedback Analysis Dashboard

Domain: College

Difficulty: ★ ★ ★ Advanced

Description:

Analyse 360-degree feedback (student → faculty, faculty → management) with quantitative scores and qualitative sentiment.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Feedback aggregator, Score chart, Sentiment display, Criterion breakdown, Trend analysis, Faculty compare, Export, Alert config, History, Admin panel

Problem 232: College Social Media Analytics

Domain: College

Difficulty: ★ ★ Intermediate

Description:

Track college social media metrics, content scheduling, engagement analytics, follower growth, and competitive benchmarking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Platform connect mock, Metrics dashboard, Content calendar, Engagement chart, Follower trend, Benchmark, Export, History, Notifications, Admin panel

Problem 233: Student Innovation & Startup Cell Portal

Domain: College

Difficulty: ★ ★ ★ Advanced

Description:

Register student startups, submit innovation proposals, apply for incubation, track mentoring, and display success stories.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Startup register, Proposal form, Incubation apply, Mentor assign, Progress tracker, Success gallery, History, Notifications, Export, Admin panel

Problem 234: Hostel Warden Management System

Domain: College

Difficulty: ★ Beginner

Description:

Manage hostel wardens, room inspection schedules, incident reports, student attendance within hostel, and guest register.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Warden profile, Inspection schedule, Incident log, Hostel attendance, Guest register, Alert config, History, Export, Notifications, Admin panel

Problem 235: College Tender & Procurement Portal

Domain: College

Difficulty: ★ ★ Intermediate

Description:

Publish and manage tenders for college construction, equipment, and services with vendor bids, evaluation, and award.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Tender post, Vendor reg, Bid submission, Evaluation form, Award notify, Contract upload, History, Export, Notifications, Admin panel

Problem 236: Online Remedial Class Scheduler

Domain: College

Difficulty: ★ ★ ★ Advanced

Description:

Identify struggling students, schedule remedial sessions, track attendance, and measure post-remedial performance improvement.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Student identify, Session form, Attendance mark, Pre/post score, Improvement chart, Faculty assign, History, Export, Notifications, Admin panel

Problem 237: College Parking & Vehicle Registration

Domain: College

Difficulty: ★ Beginner

Description:

Register student/staff vehicles, manage entry passes, parking zone allocation, violation log, and space availability display.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Vehicle form, Pass generate, Zone allocate, Violation log, Space display, History, Export, Notifications, Admin panel, Barcode mock

Problem 238: Industry-Academia Collaboration Portal

Domain: College

Difficulty: ★★ Intermediate

Description:

Manage MoUs with industry partners, internship pipelines, collaborative research, guest faculty, and industry project proposals.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

MoU register, Partner profile, Internship pipeline, Research track, Guest faculty, Project proposal, History, Export, Notifications, Admin panel

Problem 239: College Electrical & Civil Maintenance Portal

Domain: College

Difficulty: ★ Beginner

Description:

Log and track infrastructure maintenance requests — electrical, plumbing, civil — with contractor assign and completion sign-off.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Request form, Category select, Contractor assign, Status update, Completion sign-off, Cost log, History, Export, Notifications, Admin panel

Problem 240: Student Senate & SGA Portal

Domain: College

Difficulty: ★★ Intermediate

Description:

Manage student government activities, election nominations, voting, meeting minutes, resolution tracking, and budget proposals.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Election form, Nomination, Voting UI, Minutes editor, Resolution track, Budget proposal, History, Notifications, Export, Admin panel

Problem 241: College ISO Audit Management

Domain: College

Difficulty: ★ ★ ★ Advanced

Description:

Prepare and manage ISO certification audits with document control, internal audit scheduling, non-conformance tracking, and corrective actions.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Doc library, Audit schedule, NC log, Corrective action, Status track, Cert expiry alert, History, Export PDF, Notifications, Admin panel

Problem 242: Faculty Research Grant Tracker

Domain: College

Difficulty: ★ ★ Intermediate

Description:

Track research grant applications, funding agencies, awarded amounts, expenditure, interim reports, and project closure.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Grant form, Funding agency list, Award record, Expenditure log, Report upload, Closure form, History, Export, Notifications, Admin panel

Problem 243: College Emergency Notification System

Domain: College

Difficulty: ★ ★ ★ Advanced

Description:

Broadcast emergency notifications (weather, security, health) to students and staff via dashboard with acknowledgement tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Alert form, Category select, Broadcast mock, Acknowledgement track, Escalation, History, Export, Notifications, Admin panel, Priority levels

Problem 244: College Supply Chain & Stores Manager

Domain: College

Difficulty: ★ Beginner

Description:

Manage college stores inventory — stationery, lab consumables, sports equipment — with indent requests, issue records, and stock alerts.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Inventory list, Indent form, Issue record, Stock alert, Vendor order, Return log, History, Export, Notifications, Admin panel

Problem 245: Departmental Lab Equipment Booking

Domain: College

Difficulty: ★ ★ Intermediate

Description:

Book lab equipment and instruments for experiments, research, and practical sessions with usage logs and maintenance scheduling.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Equipment list, Booking form, Usage log, Maintenance schedule, Availability display, Conflict check, History, Export, Notifications, Admin panel

Problem 246: College Diversity & Inclusion Dashboard

Domain: College

Difficulty: ★ ★ ★ Advanced

Description:

Track diversity metrics across gender, category, and disability status in admissions, faculty recruitment, and scholarships.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Diversity chart, Metric input, Trend line, Department breakdown, Scholarship track, Policy display, Export, History, Notifications, Admin panel

Problem 247: Student Union Election Portal

Domain: College

Difficulty: ★ Beginner

Description:

End-to-end student union election management with candidate nominations, voter registration, online voting, result declaration, and audit log.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Nomination form, Voter register, Voting UI, Result tally, Audit log, Announcement, History, Export, Notifications, Admin panel



Bank Domain

This section contains 50 complete hackathon problem statements for the Bank Domain. Each problem is ready for immediate prototype development.

Problem 248: Online Account Opening Portal

Domain: Bank

Difficulty: ★ Beginner

Description:

Digital bank account opening with KYC document upload, video verification mock, account type selection, and instant account number generation.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Application form, KYC upload, Video KYC mock, Account type, Instant number, Welcome kit mock, Status track, Notifications, Admin panel, Postman API

Problem 249: Internet Banking Dashboard

Domain: Bank

Difficulty: ★ ★ Intermediate

Description:

Comprehensive internet banking UI with account summary, recent transactions, quick transfer, bills, FD/RD, and profile management.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Account summary, Transaction list, Quick transfer, Bill pay, FD/RD summary, Profile settings, Alerts, Export statement, Notifications, Admin panel

Problem 250: Fund Transfer System (NEFT/RTGS/IMPS)

Domain: Bank

Difficulty: ★ ★ ★ Advanced

Description:

Transfer funds between accounts with beneficiary management, transfer limits, transaction OTP mock, status tracking, and receipt generation.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Beneficiary add, Transfer form, Limit display, OTP mock, Status tracker, Receipt PDF, Transaction history, Notifications, Admin panel, Postman API

Problem 251: Loan Application Portal

Domain: Bank

Difficulty: ★★ Intermediate

Description:

Apply for home, personal, vehicle, or education loans with eligibility calculator, document upload, EMI preview, and status tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Loan type select, Eligibility calc, Application form, Doc upload, EMI preview, Status tracker, Offer display, Sanction letter PDF, Notifications, Admin panel

Problem 252: Credit Card Management Portal

Domain: Bank

Difficulty: ★★★ Advanced

Description:

Manage credit cards — view statements, make payments, set limits, report lost/stolen, convert to EMI, and track rewards points.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Statement view, Payment form, Limit setter, Block card, EMI convert, Rewards tracker, History, Export, Notifications, Admin panel

Problem 253: Fixed Deposit & Recurring Deposit Manager

Domain: Bank

Difficulty: ★ Beginner

Description:

Open, track, renew, and prematurely close FDs and RDs with interest calculation, maturity alerts, and certificate generation.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

FD/RD form, Interest calc, Maturity alert, Renewal flow, Premature close, Certificate PDF, History, Notifications, Admin panel, Postman API

Problem 254: Bank Branch Locator & Appointment Scheduler

Domain: Bank

Difficulty: ★★ Intermediate

Description:

Find nearby branches and ATMs with service filters, book in-branch appointments for specific services, and receive slot confirmations.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Branch search, Map view, Service filter, Slot picker, Appointment form, Confirm notify, History, Admin panel, Postman API, Export

Problem 255: KYC & AML Compliance Dashboard

Domain: Bank

Difficulty: ★★★ Advanced

Description:

Monitor customer KYC expiry, re-KYC workflows, AML alert management, risk scoring, and regulatory reporting mock.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

KYC status list, Expiry alert, Re-KYC form, AML alert queue, Risk score display, Regulatory report, History, Export, Notifications, Admin panel

Problem 256: Cheque Book & Demand Draft Request

Domain: Bank

Difficulty: ★ Beginner

Description:

Request cheque books and demand drafts online with delivery address, DD beneficiary details, charges display, and status tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Cheque book form, DD form, Delivery address, Charges display, Status tracker, Dispatch notify, History, Export, Notifications, Admin panel

Problem 257: Bank Statement Download Portal

Domain: Bank

Difficulty: ★★ Intermediate

Description:

Generate and download account statements for custom date ranges in PDF/CSV with transaction filtering and e-statement subscription.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Date range picker, Format select, Filter panel, Generate button, Download link, E-statement sub, History, Notifications, Admin panel, Postman API

Problem 258: Debit Card Management Portal

Domain: Bank

Difficulty: ★ Beginner

Description:

Manage debit cards — activate, block, change PIN mock, set international/online usage limits, and view transaction alerts.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Card list, Activate form, Block flow, PIN change mock, Usage limits, Alert settings, History, Export, Notifications, Admin panel

Problem 259: UPI & Mobile Banking Registration

Domain: Bank

Difficulty: ★ ★ Intermediate

Description:

Register and manage UPI IDs, link bank accounts, set transaction limits, manage mandates, and view UPI transaction history.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

UPI registration, Account link, Limit setter, Mandate manage, Transaction history, QR code display, Notifications, Export, Admin panel, Postman API

Problem 260: Insurance Premium Payment Portal

Domain: Bank

Difficulty: ★ ★ ★ Advanced

Description:

Pay life, health, vehicle, and property insurance premiums via internet banking with policy search, payment confirmation, and receipt.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Policy search, Premium display, Payment form, Receipt PDF, Policy history, Reminders, Notifications, Admin panel, Postman API, Export

Problem 261: Mutual Fund Investment Portal

Domain: Bank

Difficulty: ★ ★ Intermediate

Description:

Browse, compare, and invest in mutual funds with SIP setup, portfolio dashboard, NAV tracker, redemption flow, and performance charts.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Fund catalogue, Compare tool, SIP form, Portfolio dash, NAV tracker, Redemption form, Performance chart, History, Notifications, Admin panel

Problem 262: Bank Grievance & Complaint System

Domain: Bank

Difficulty: ★ ★ ★ Advanced

Description:

Submit banking complaints (wrong debit, fraud, service issues) with complaint ID, escalation to ombudsman, and resolution tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Complaint form, Category select, Status tracker, Escalation flow, Ombudsman link, Resolution note, History, Export, Notifications, Admin panel

Problem 263: NRI Banking Services Portal

Domain: Bank

Difficulty: ★ Beginner

Description:

Manage NRE/NRO accounts, foreign remittance, FCNR deposits, repatriation requests, and tax documents for non-resident Indians.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Account summary, Remittance form, FCNR manage, Repatriation form, Tax doc download, Exchange rate, History, Export, Notifications, Admin panel

Problem 264: Bank ATM Network Management UI

Domain: Bank

Difficulty: ★★ Intermediate

Description:

Monitor ATM uptime, cash levels, fault logs, replenishment schedules, and service requests across the bank's ATM network.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

ATM map, Status display, Cash level alert, Fault log, Replenishment form, Service assign, History, Export, Notifications, Admin panel

Problem 265: Corporate Banking Trade Finance Portal

Domain: Bank

Difficulty: ★★★ Advanced

Description:

Manage letters of credit, bank guarantees, bill discounting, export collections, and trade document submission for corporate clients.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

LC application, BG form, Bill discounting, Export collection, Document upload, Status tracker, History, Export PDF, Notifications, Admin panel

Problem 266: Wealth Management Advisory UI

Domain: Bank

Difficulty: ★ Beginner

Description:

Personalised investment advisory with risk profiling, asset allocation recommendations, portfolio rebalancing, and goal-based planning.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Risk profile form, Asset allocation chart, Recommendation display, Portfolio view, Rebalance suggest, Goal planner, History, Export, Notifications, Admin panel

Problem 267: Bank Fraud Detection Alert Dashboard

Domain: Bank

Difficulty: ★★ Intermediate

Description:

Real-time dashboard for fraud analysts to review flagged transactions, mark genuine/fraud, block cards, and generate incident reports.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Alert queue, Transaction detail, Flag display, Action buttons, Block card, Incident report, History, Export, Notifications, Admin panel

Problem 268: Locker Services Management Portal

Domain: Bank

Difficulty: ★ Beginner

Description:

Apply for bank locker, manage access schedules, log entries, pay locker rent, and view locker agreement.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Locker application, Size select, Rent payment, Access log, Agreement view, Reminder, History, Export, Notifications, Admin panel

Problem 269: Bank Payroll Services Dashboard

Domain: Bank

Difficulty: ★★ Intermediate

Description:

Corporate clients process employee salary disbursements, manage payroll accounts, upload salary files, and view disbursement reports.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Payroll upload, Account list, Salary preview, Disburse flow, Report download, Error log, History, Export, Notifications, Admin panel

Problem 270: Credit Score & CIBIL Tracker

Domain: Bank

Difficulty: ★★ ★ Advanced

Description:

Check credit score, view CIBIL report breakdown, dispute errors, track score improvement over time, and get loan eligibility hints.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Score display, Report view, Dispute form, Score trend chart, Improvement tips, Eligibility hint, History, Notifications, Admin panel, Export

Problem 271: Bank Auction & Recovery Portal

Domain: Bank

Difficulty: ★★ ★ Intermediate

Description:

Manage non-performing asset auctions — property listings, bid submission, payment collection, and legal notice tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Property list, Bid form, Payment mock, Legal notice, Status tracker, Document upload, History, Export, Notifications, Admin panel

Problem 272: Banking Chatbot UI Mock

Domain: Bank

Difficulty: ★★ ★ Advanced

Description:

Conversational banking assistant UI for balance enquiries, mini-statement, fund transfer initiation, and FAQs with escalation to human agent.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Chat UI, Intent display, Balance mock, Transfer initiate, FAQ display, Escalate option, History, Profile, Notifications, Admin panel

Problem 273: Branch Performance Dashboard

Domain: Bank

Difficulty: ★ Beginner

Description:

Track branch-level KPIs — account openings, loan disbursements, CASA ratio, cross-sell metrics — with daily/monthly trends.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

KPI cards, Trend charts, Branch compare, Loan disbursement, CASA chart, Cross-sell display, Date filter, Export, Notifications, Admin panel

Problem 274: Bank Regulatory Reporting UI

Domain: Bank

Difficulty: ★★ Intermediate

Description:

Prepare and submit regulatory reports (Basel III, CRAR, SLR, CRR) with data input, calculation display, and submission tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Report type select, Data entry form, Calculation display, Validation check, Submit flow, Status tracker, History, Export PDF, Notifications, Admin panel

Problem 275: Bank Reconciliation Tool

Domain: Bank

Difficulty: ★★★ Advanced

Description:

Reconcile bank statements with internal ledger entries, flag unmatched transactions, mark resolved, and generate reconciliation report.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Statement upload, Ledger import, Match engine, Flag unmatched, Mark resolved, Report PDF, History, Export, Notifications, Admin panel

Problem 276: Bank Staff Training & LMS Portal

Domain: Bank

Difficulty: ★ Beginner

Description:

Assign product, compliance, and soft skills training to bank staff with quiz, score tracking, certification, and completion reports.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Training catalogue, Module assign, Video/quiz, Score display, Certificate, Progress tracker, History, Export, Notifications, Admin panel

Problem 277: Digital Onboarding for SME Banking

Domain: Bank

Difficulty: ★★ Intermediate

Description:

Onboard small businesses for current accounts, business loans, and payment solutions with GST/CIN verification and KYC workflow.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Business form, GST/CIN verify mock, KYC upload, Account type, Loan apply, Payment solution, Status track, Notifications, Admin panel, Postman API

Problem 278: Bank Yield & Interest Rate Dashboard

Domain: Bank

Difficulty: ★ Beginner

Description:

Display current deposit and lending rates, rate change history, RBI policy tracker, and competitor rate comparison.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Rate table, Rate history chart, RBI tracker, Competitor compare, Alert config, Export, History, Notifications, Admin panel, Search

Problem 279: Bill Payment & Recharge Portal

Domain: Bank

Difficulty: ★★ Intermediate

Description:

Pay utility bills, mobile recharges, DTH, OTT subscriptions, and municipal taxes with biller registration, payment history, and auto-pay setup.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Biller list, Payment form, Auto-pay setup, Payment history, Receipt PDF, Reminder, Notifications, Admin panel, Postman API, Export

Problem 280: Bank Network API Testing Dashboard

Domain: Bank

Difficulty: ★ ★ ★ Advanced

Description:

Internal Postman-style tool for bank developers to test payment APIs (NEFT, IMPS, UPI) with authentication, request builder, and response logs.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Request builder, Auth header, Environment select, Response viewer, Test suite, History, Mock server, Error log, Export, Docs panel

Problem 281: Green Banking & Sustainability Tracker

Domain: Bank

Difficulty: ★ ★ Intermediate

Description:

Track bank's sustainability initiatives — paperless accounts, green loans, carbon footprint reduction, and ESG score display.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Sustainability KPIs, Green loan tracker, Carbon chart, Paperless adoption, ESG score, Initiative list, History, Export, Notifications, Admin panel

Problem 282: Bank Customer 360 CRM View

Domain: Bank

Difficulty: ★ ★ ★ Advanced

Description:

Unified customer view for relationship managers with account portfolio, interactions, products held, service requests, and next-best-action.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Customer summary, Product held, Interaction log, Service request, NBA display, Note add, Profile, History, Export, Admin panel

Problem 283: Bank Holiday & Service Calendar

Domain: Bank

Difficulty: ★ Beginner

Description:

Publish and manage bank holidays, service downtime schedules, and branch operational hours with customer notification.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Holiday list, Add form, Region filter, Service downtime, Branch hours, Notify customers, History, Export, Notifications, Admin panel

Problem 284: Bank Internal Audit Management

Domain: Bank

Difficulty: ★ ★ Intermediate

Description:

Plan and execute internal audits across branches with risk-based audit scheduling, finding log, management response, and closure tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Audit schedule, Branch assign, Finding log, Risk rating, Management response, Closure form, History, Export, Notifications, Admin panel

Problem 285: Bank Succession & Emergency Contact Mandate

Domain: Bank

Difficulty: ★ ★ ★ Advanced

Description:

Register nominees, update emergency contacts, manage mandate holders, and set transfer instructions for accounts.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Nominee form, Emergency contact, Mandate add, Transfer instruction, Verification mock, Status track, History, Export, Notifications, Admin panel

Problem 286: Bank Trade Repository UI

Domain: Bank

Difficulty: ★ Beginner

Description:

Log and view OTC derivative transactions, trade confirmations, valuation, margin calls, and regulatory trade reporting.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Trade log, Confirmation view, Valuation display, Margin call alert, Regulatory report, History, Export, Notifications, Admin panel, Postman API

Problem 287: Bank Treasury Operations Dashboard

Domain: Bank

Difficulty: ★★ Intermediate

Description:

Treasury management UI for monitoring money market deals, forex positions, investment portfolio, and liquidity ratios.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Deal log, Forex position, Investment portfolio, Liquidity ratio, Alert config, Date filter, Export, History, Notifications, Admin panel

Problem 288: Bank Mergers & Acquisitions Due Diligence Tool

Domain: Bank

Difficulty: ★ Beginner

Description:

Organise M&A due diligence documents, assign reviewer teams, track review progress, issue risk flags, and generate DD report.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Document upload, Reviewer assign, Progress tracker, Risk flag, Review notes, DD report PDF, History, Export, Notifications, Admin panel

Problem 289: Bank Whistleblower Portal

Domain: Bank

Difficulty: ★★ Intermediate

Description:

Anonymous channel for reporting banking malpractice, fraud, and misconduct with case ID, secure submission, and investigation tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Report form, Anonymity option, Case ID generate, Status tracker (anonymous), Investigation assign, Outcome notify, History, Export, Notifications, Admin panel

Problem 290: Bank Penalty & Charges Calculator

Domain: Bank

Difficulty: ★ ★ ★ Advanced

Description:

Calculate banking penalties — late EMI, prepayment, locker rent, minimum balance — with applicable fee tables and waiver requests.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Penalty type select, Input form, Calc result, Fee table, Waiver request form, History, Export, Notifications, Admin panel, Postman API

Problem 291: Bank Customer Risk Profiling Tool

Domain: Bank

Difficulty: ★ ★ Intermediate

Description:

KYC-linked risk profiling for investment suitability, AML risk categorisation, periodic review, and risk-based monitoring triggers.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Profile form, Risk score display, Category badge, Review scheduler, Monitoring trigger, History, Export, Notifications, Admin panel, Postman API

Problem 292: Bank Procurement & Vendor Management Portal

Domain: Bank

Difficulty: ★ ★ ★ Advanced

Description:

Manage vendor empanelment, RFP process, contract management, SLA monitoring, and payment processing for bank suppliers.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Vendor register, RFP form, Bid compare, Contract upload, SLA tracker, Payment log, History, Export, Notifications, Admin panel

Problem 293: Bank Annual Report Builder

Domain: Bank

Difficulty: ★ Beginner

Description:

Compile and generate bank annual report sections — financial highlights, MD&A, risk disclosures — from structured inputs with chart generation.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Section editor, Chart generator, Financial input, MD&A editor, Risk table, PDF renderer, Version control, History, Export, Notifications

Problem 294: Bank Investor Relations Portal

Domain: Bank

Difficulty: ★ ★ Intermediate

Description:

Publish quarterly results, analyst presentations, regulatory filings, AGM notices, and shareholder meeting schedules.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Filing list, Result upload, Presentation link, AGM notice, Schedule display, Subscribe mock, Search, History, Notifications, Admin panel

Problem 295: MSME Loan Processing Dashboard

Domain: Bank

Difficulty: ★ ★ ★ Advanced

Description:

End-to-end MSME loan processing from application to disbursement with credit appraisal, committee approval, and disbursement tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Application form, Credit appraisal, Committee approve, Sanction letter, Disburse flow, Status track, History, Export, Notifications, Admin panel

Problem 296: Bank Security Incident Response Dashboard

Domain: Bank

Difficulty: ★ Beginner

Description:

Cybersecurity incident management for banks with threat alerts, incident classification, response actions, and forensic report tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Alert queue, Incident form, Threat classify, Response checklist, Forensic report, Status track, History, Export, Notifications, Admin panel

Problem 297: Bank Customer Lifetime Value Dashboard

Domain: Bank

Difficulty: ★★ Intermediate

Description:

Analyse and visualise CLV, churn risk, product penetration, and campaign performance for customer segments.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

CLV chart, Churn risk table, Product penetration, Campaign performance, Segment filter, Date range, Export, Notifications, Admin panel, Postman API

Online Shopping Apps Domain

This section contains 50 complete hackathon problem statements for the Online Shopping Apps Domain. Each problem is ready for immediate prototype development.

Problem 298: E-Commerce Product Catalogue

Domain: Online Shopping Apps

Difficulty: ★ Beginner

Description:

Browse products with multi-level categories, rich filters (price, brand, rating, colour, size), search suggestions, and comparison.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Product grid, Category nav, Filter panel, Search bar, Sort options, Compare tool, Product detail, Wishlist, History, Admin panel

Problem 299: Shopping Cart & Checkout Flow

Domain: Online Shopping Apps

Difficulty: ★ ★ Intermediate

Description:

Multi-step checkout with cart management, address selection, coupon application, payment method selection, and order confirmation.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Cart UI, Address form, Coupon input, Payment selector, Order summary, Confirm screen, Email mock, History, Notifications, Admin panel

Problem 300: User Registration & Profile Management

Domain: Online Shopping Apps

Difficulty: ★ ★ ★ Advanced

Description:

Register, login, manage profile, address book, order history, wishlist, and account security settings for shoppers.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Signup/login form, Profile edit, Address book, Order history, Wishlist, Password change, 2FA mock, Notifications, Export, Admin panel

Problem 301: Product Review & Rating System

Domain: Online Shopping Apps

Difficulty: ★★ Intermediate

Description:

Submit verified purchase reviews with star ratings, photo uploads, helpful votes, Q&A, and seller response integration.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Review form, Star rating, Photo upload, Helpful vote, Q&A section, Seller response, Review list, Filter, History, Admin panel

Problem 302: Order Tracking & Delivery Status

Domain: Online Shopping Apps

Difficulty: ★★★ Advanced

Description:

Real-time order tracking with shipment milestones, courier partner mock, estimated delivery, delay alerts, and delivery proof.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Tracking number input, Milestone timeline, Map mock, Courier display, Delay alert, Proof upload, History, Notifications, Admin panel, Export

Problem 303: Flash Sale & Deals Engine

Domain: Online Shopping Apps

Difficulty: ★ Beginner

Description:

Create and display time-limited flash sales with countdown timers, stock counters, deal categories, and personalised deal recommendations.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Deal catalogue, Countdown timer, Stock bar, Category filter, Personalised deals, Notify me, Cart link, History, Notifications, Admin panel

Problem 304: Product Return & Refund Portal

Domain: Online Shopping Apps

Difficulty: ★★ Intermediate

Description:

Initiate returns/exchanges with reason selection, photo upload, pick-up scheduling, refund method selection, and status tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Return form, Reason select, Photo upload, Pickup schedule, Refund method, Status tracker, History, Email notify, Admin panel, Postman API

Problem 305: Seller Onboarding & Dashboard

Domain: Online Shopping Apps

Difficulty: ★★★ Advanced

Description:

Onboard marketplace sellers with business verification, product listing, pricing management, order fulfilment, and revenue dashboard.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Seller form, Doc upload, Product add, Pricing manager, Order queue, Revenue chart, History, Notifications, Export, Admin panel

Problem 306: Shopping Wishlist & Save for Later

Domain: Online Shopping Apps

Difficulty: ★ Beginner

Description:

Create and manage multiple wishlists, share wishlists, price drop alerts, stock alerts, and move items between wishlist and cart.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Wishlist create, Item add, Multiple lists, Share link, Price alert, Stock alert, Cart move, History, Notifications, Admin panel

Problem 307: Product Bundle & Kit Builder

Domain: Online Shopping Apps

Difficulty: ★★ Intermediate

Description:

Create custom product bundles and gift kits with individual item selection, pricing, gifting message, packaging options, and preview.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Item selector, Bundle price calc, Gift message, Packaging option, Preview, Add to cart, History, Notifications, Admin panel, Postman API

Problem 308: Subscription Box Management

Domain: Online Shopping Apps

Difficulty: ★ Beginner

Description:

Manage recurring product subscription boxes with customisation, skip/pause/cancel, billing management, and delivery schedule.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Subscription form, Box customise, Billing manage, Skip/pause/cancel, Delivery schedule, History, Notifications, Export, Admin panel, Postman API

Problem 309: Loyalty Points & Rewards Programme

Domain: Online Shopping Apps

Difficulty: ★ ★ Intermediate

Description:

Earn and redeem points on purchases with tier benefits, point expiry, referral bonuses, and cashback conversion.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Points dashboard, Tier display, Earn history, Redeem flow, Referral link, Expiry alert, Cashback convert, History, Notifications, Admin panel

Problem 310: Product Search & Recommendation Engine UI

Domain: Online Shopping Apps

Difficulty: ★ ★ ★ Advanced

Description:

Advanced product search with voice input mock, image search mock, autocomplete, related products, and personalised recommendations.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Search bar, Voice mock, Image search mock, Autocomplete, Related products, Recommendation grid, History, Profile link, Admin panel, Postman API

Problem 311: Grocery & Fresh Produce Ordering

Domain: Online Shopping Apps

Difficulty: ★ ★ Intermediate

Description:

Order groceries and fresh produce with category-wise browsing, substitution preferences, delivery slot selection, and recurring order.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Category grid, Substitution pref, Delivery slot, Recurring order, Cart, Checkout, Tracking mock, History, Notifications, Admin panel

Problem 312: Fashion Lookbook & Style Guide

Domain: Online Shopping Apps

Difficulty: ★ ★ ★ Advanced

Description:

Interactive fashion lookbook with outfit builder, style tips, shop-the-look links, seasonal collections, and personalised styling.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Lookbook grid, Outfit builder, Shop-the-look, Season filter, Personalise quiz, Style tips, Wishlist, History, Notifications, Admin panel

Problem 313: Electronics Comparison & Spec Checker

Domain: Online Shopping Apps

Difficulty: ★ Beginner

Description:

Compare electronics specs side-by-side with compatibility checker, expert review links, price history chart, and buy decision helper.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Product compare, Spec table, Compatibility check, Price history, Expert reviews, Decision helper, Cart link, History, Notifications, Admin panel

Problem 314: Shopping App Onboarding & Preference Setup

Domain: Online Shopping Apps

Difficulty: ★★ Intermediate

Description:

New user onboarding with interest selection, size/preference inputs, location setup, personalised home feed, and push notification opt-in.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Onboarding steps, Interest picker, Size input, Location setup, Feed preview, Notification opt-in, Profile, History, Admin panel, Notifications

Problem 315: Coupon & Promo Code Manager

Domain: Online Shopping Apps

Difficulty: ★★★ Advanced

Description:

Create, manage, and track promotional codes with discount type, usage limits, expiry, customer segment targeting, and redemption analytics.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Coupon form, Discount type, Usage limit, Expiry date, Segment target, Redemption chart, History, Export, Notifications, Admin panel

Problem 316: Buy Now Pay Later (BNPL) Integration UI

Domain: Online Shopping Apps

Difficulty: ★ Beginner

Description:

Display BNPL options at checkout with eligibility check mock, instalment plans, interest display, and repayment schedule.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

BNPL option display, Eligibility mock, Plan select, Interest display, Repayment schedule, Accept terms, History, Notifications, Admin panel, Postman API

Problem 317: Product Inventory & Warehouse Dashboard

Domain: Online Shopping Apps

Difficulty: ★★ Intermediate

Description:

Track stock levels across warehouses, manage reorder points, view incoming shipments, and generate inventory reports.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Inventory table, Warehouse filter, Reorder alert, Shipment log, Stock chart, Export, History, Notifications, Admin panel, Postman API

Problem 318: Customer Support Ticket System

Domain: Online Shopping Apps

Difficulty: ★ Beginner

Description:

Shopping app support portal for order issues, product queries, and account problems with live chat mock, ticket tracking, and resolution.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Ticket form, Category select, Chat mock, Status track, Resolution note, Satisfaction score, History, Export, Notifications, Admin panel

Problem 319: Multi-Vendor Marketplace Admin Panel

Domain: Online Shopping Apps

Difficulty: ★★ Intermediate

Description:

Admin dashboard for managing sellers, product approvals, commission rates, dispute resolution, and platform-wide analytics.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Seller list, Product approval, Commission config, Dispute queue, Analytics dashboard, Export, Notifications, History, Admin tools, Postman API

Problem 320: Augmented Reality Product Preview UI

Domain: Online Shopping Apps

Difficulty: ★★★ Advanced

Description:

Mock AR try-on interface for fashion, furniture, and accessories with camera feed overlay, product switcher, and share feature.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

AR canvas mock, Camera feed, Product overlay, Switch product, Size adjust, Capture, Share, History, Notifications, Admin panel

Problem 321: Gift Card & Voucher System

Domain: Online Shopping Apps

Difficulty: ★ ★ Intermediate

Description:

Purchase, send, and redeem digital gift cards and vouchers with balance display, partial redemption, transfer, and expiry tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Gift card purchase, Send form, Redeem input, Balance display, Transfer, Expiry alert, History, Notifications, Admin panel, Postman API

Problem 322: Product Q&A Community Forum

Domain: Online Shopping Apps

Difficulty: ★ ★ ★ Advanced

Description:

Community-driven Q&A on product pages with upvotes, seller verification badges, follow question, and notification subscriptions.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Q&A form, Upvote, Seller badge, Follow question, Search, Filter, Notification, Profile, History, Admin panel

Problem 323: Shopping Analytics Dashboard for Sellers

Domain: Online Shopping Apps

Difficulty: ★ Beginner

Description:

Seller analytics on sales volume, conversion rate, return rate, customer demographics, and top products with actionable insights.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Sales chart, Conversion rate, Return rate, Demographics, Top products, Insight cards, Date filter, Export, Notifications, Admin panel

Problem 324: Same-Day Delivery Slot Booking

Domain: Online Shopping Apps

Difficulty: ★ ★ Intermediate

Description:

Book same-day or next-day delivery slots by pin code with time window selection, live capacity display, and rescheduling option.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Slot calendar, Pin code check, Time window, Capacity display, Reschedule, Confirm notify, History, Notifications, Admin panel, Postman API

Problem 325: Charity & Round-Up Donation Feature

Domain: Online Shopping Apps

Difficulty: ★ ★ ★ Advanced

Description:

Round up purchase price to donate micro-amounts to charity, select NGOs, track total donated, and generate donation certificate.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Round-up toggle, NGO select, Amount display, Donation history, Total donated, Certificate PDF, Profile, Notifications, Admin panel, Export

Problem 326: Product Authenticity Verification UI

Domain: Online Shopping Apps

Difficulty: ★ Beginner

Description:

Upload product serial number or QR code to verify authenticity with brand database mock, status display, and report counterfeit.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

QR/serial input, Verify mock, Status display, Brand info, Counterfeit report form, History, Notifications, Admin panel, Postman API, Export

Problem 327: Shopping Chatbot & Virtual Assistant UI

Domain: Online Shopping Apps

Difficulty: ★★ Intermediate

Description:

AI shopping assistant interface for product recommendations, order help, and FAQ with intent classification mock and escalation.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Chat UI, Recommendation display, Order help, FAQ, Intent mock, Escalate option, History, Profile, Notifications, Admin panel

Problem 328: Hyperlocal Shop & Delivery App UI

Domain: Online Shopping Apps

Difficulty: ★ Beginner

Description:

Browse local shops near the user, view real-time stock, order, and track hyperlocal delivery within hours.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Shop list, Map view, Product browse, Cart, Delivery tracker, ETA display, History, Notifications, Admin panel, Postman API

Problem 329: B2B Wholesale Ordering Portal

Domain: Online Shopping Apps

Difficulty: ★★ Intermediate

Description:

Bulk order portal for business buyers with MOQ display, tiered pricing, credit limit, purchase order upload, and invoice management.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Product catalogue, MOQ display, Tiered pricing, Credit limit, PO upload, Invoice manage, History, Export, Notifications, Admin panel

Problem 330: Product Recall & Safety Alert System

Domain: Online Shopping Apps

Difficulty: ★★ ★ Advanced

Description:

Notify customers about recalled products with safety details, replacement/refund options, and regulatory compliance tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Recall alert, Product identify, Safety details, Replacement form, Refund option, Compliance log, History, Export, Notifications, Admin panel

Problem 331: Personalised Email & Push Campaign Builder

Domain: Online Shopping Apps

Difficulty: ★★ Intermediate

Description:

Create targeted marketing campaigns with user segment selection, A/B test config, preview, schedule, and performance tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Campaign form, Segment select, A/B test, Preview, Schedule, Send mock, Performance chart, History, Export, Admin panel

Problem 332: Size Guide & Fit Recommender

Domain: Online Shopping Apps

Difficulty: ★★ ★ Advanced

Description:

Interactive size guide with body measurement input, brand-specific size conversion, fit type selection, and size recommendation.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Measurement form, Brand converter, Fit type, Size recommend, Guide table, Save measurements, History, Notifications, Admin panel, Profile

Problem 333: Product Video & 360° Image Viewer

Domain: Online Shopping Apps

Difficulty: ★ Beginner

Description:

Rich product media viewer with video playback, 360° rotation, zoom, and hotspot annotations pointing to features.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Media gallery, Video player, 360° canvas, Zoom control, Hotspot display, Share, Wishlist link, History, Admin panel, Notifications

Problem 334: Influencer Affiliate Programme Portal

Domain: Online Shopping Apps

Difficulty: ★ ★ Intermediate

Description:

Influencers join the programme, get unique links, track clicks/conversions, view commissions, and request payouts.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Registration form, Link generator, Click tracker, Conversion chart, Commission display, Payout request, History, Notifications, Export, Admin panel

Problem 335: Age-Gated Product Category UI

Domain: Online Shopping Apps

Difficulty: ★ ★ ★ Advanced

Description:

Age verification and parental controls for restricted product categories with DOB verification, guardian consent, and access management.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Age verification form, DOB check, Guardian consent, Category lock, Access log, History, Notifications, Admin panel, Export, Postman API

Problem 336: Abandoned Cart Recovery UI

Domain: Online Shopping Apps

Difficulty: ★ Beginner

Description:

Track and display abandoned carts, trigger reminder notifications, apply auto-discount, and measure recovery rate.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Abandoned list, Item preview, Reminder trigger, Discount apply, Recovery rate chart, History, Notifications, Admin panel, Export, Postman API

Problem 337: Shopping Super App Integration Hub

Domain: Online Shopping Apps

Difficulty: ★ ★ Intermediate

Description:

Integrate shopping, travel, food delivery, and financial services in a super-app UI with unified wallet, history, and notifications.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Service tiles, Unified wallet, Cross-service history, Notifications centre, Profile, Search, Reward aggregator, Settings, Admin panel, Postman API

Problem 338: Reverse Logistics & Bulk Return Manager

Domain: Online Shopping Apps

Difficulty: ★ Beginner

Description:

Manage bulk returns from corporate clients or marketplace sellers with return manifests, quality inspection, restock, and credit notes.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Return manifest, Bulk input, Inspection checklist, Restock update, Credit note generate, Status track, History, Export, Notifications, Admin panel

Problem 339: Rental & Sharing Economy Product Portal

Domain: Online Shopping Apps

Difficulty: ★ ★ Intermediate

Description:

Rent products (electronics, furniture, tools) with rental duration, deposit calculation, delivery, condition report, and return workflow.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Rental catalogue, Duration picker, Deposit calc, Delivery form, Condition report, Return flow, History, Notifications, Admin panel, Postman API

Problem 340: Eco-Friendly & Sustainable Products Badge System

Domain: Online Shopping Apps

Difficulty: ★ ★ ★ Advanced

Description:

Certify and display eco-scores for products with sustainability criteria, brand commitment tracking, and green purchase incentives.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Eco-score display, Criteria checklist, Brand badge, Green incentives, Filter eco products, Impact chart, History, Notifications, Admin panel, Export

Problem 341: Live Commerce & Video Shopping UI

Domain: Online Shopping Apps

Difficulty: ★ ★ Intermediate

Description:

Live shopping stream interface with product showcase, real-time viewer count, in-stream purchase, comment feed, and replay.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Stream player mock, Product cards, In-stream cart, Comment feed, Viewer count, Replay, Like button, Profile, History, Admin panel

Problem 342: Dynamic Pricing & Price Drop Alert

Domain: Online Shopping Apps

Difficulty: ★ ★ ★ Advanced

Description:

Set price drop alerts for wishlisted products, view price history charts, track competitor prices, and receive push notifications.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Alert form, Price history chart, Competitor compare, Push notify mock, Wishlist link, Alert manage, History, Export, Admin panel, Notifications

Problem 343: International Shipping & Customs Estimator

Domain: Online Shopping Apps

Difficulty: ★ Beginner

Description:

Calculate international shipping costs, estimated duties, delivery timelines, and restricted item checks for cross-border orders.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Country select, Weight/dim input, Cost calc, Duty estimate, Delivery timeline, Restricted check, History, Export, Notifications, Admin panel

Problem 344: Corporate Gifting Portal

Domain: Online Shopping Apps

Difficulty: ★ ★ Intermediate

Description:

Order bulk corporate gifts with branding customisation, company invoice, bulk delivery management, and recipient tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Product select, Branding input, Bulk order form, Company invoice, Delivery manage, Recipient track, History, Export, Notifications, Admin panel

Problem 345: Shopping App Dark Mode & Accessibility

Domain: Online Shopping Apps

Difficulty: ★ ★ ★ Advanced

Description:

Implement dark mode toggle, font size controls, high contrast, screen reader hints, and keyboard navigation for shopping apps.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Dark mode toggle, Font size slider, Contrast toggle, SR hints, Keyboard nav, Preview, Profile save, History, Admin panel, Export

Problem 346: Price Negotiation & Best Offer System

Domain: Online Shopping Apps

Difficulty: ★ Beginner

Description:

Allow buyers to submit best-offer bids on eligible products with auto-accept/reject rules, counter-offer, and negotiation history.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Eligible product flag, Offer form, Auto-rule config, Counter-offer, Acceptance notify, Negotiation history, Profile, Notifications, Admin panel, Postman API

Problem 347: Shopping Super Loyalty Cashback Dashboard

Domain: Online Shopping Apps

Difficulty: ★ ★ Intermediate

Description:

Cashback programme UI with transaction-level cashback calculation, wallet credit tracking, tier multipliers, and redemption flow.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Cashback dashboard, Transaction list, Tier multiplier, Wallet balance, Redeem flow, History, Export, Notifications, Profile, Admin panel

Meeseva (Gov Services) Domain

This section contains 50 complete hackathon problem statements for the Meeseva (Gov Services) Domain. Each problem is ready for immediate prototype development.

Problem 348: Meeseva Citizen Registration Portal

Domain: Meeseva (Gov Services)

Difficulty: ★ Beginner

Description:

Register citizens for Meeseva services with Aadhaar mock verification, address proof upload, mobile OTP, and service profile creation.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Registration form, Aadhaar mock, OTP verify, Address upload, Profile create, Status track, Notifications, Export, Admin panel, Postman API

Problem 349: Birth Certificate Application System

Domain: Meeseva (Gov Services)

Difficulty: ★ ★ Intermediate

Description:

Apply for birth certificates with hospital record input, document upload, payment mock, verification, and certificate download.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Application form, Hospital record, Doc upload, Payment mock, Verification step, Certificate PDF, Status track, Notifications, Admin panel, Export

Problem 350: Death Certificate Application Portal

Domain: Meeseva (Gov Services)

Difficulty: ★ ★ ★ Advanced

Description:

Submit death certificate applications with medical record upload, applicant relationship verification, fee payment, and issuance.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Application form, Medical doc upload, Relationship verify, Payment mock, Certificate PDF, Status track, Notifications, Admin panel, Export, Postman API

Problem 351: Caste Certificate Application System

Domain: Meeseva (Gov Services)

Difficulty: ★★ Intermediate

Description:

Apply for caste/community certificates with supporting documents, revenue authority routing, status tracking, and certificate download.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Application form, Doc upload, Revenue routing, Status tracker, Certificate PDF, Appeal form, Notifications, Admin panel, Export, Postman API

Problem 352: Income Certificate Application Portal

Domain: Meeseva (Gov Services)

Difficulty: ★★★ Advanced

Description:

Apply for income certificates with income proof upload, self-declaration, officer review, and digitally signed certificate generation.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Application form, Income proof upload, Self-declaration, Officer review, Digital cert PDF, Status track, Notifications, Admin panel, Export, Postman API

Problem 353: Residence Certificate Application

Domain: Meeseva (Gov Services)

Difficulty: ★ Beginner

Description:

Apply for domicile/residence certificates with utility bill proof, address verification, approval workflow, and digital issuance.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Application form, Address proof, Utility bill upload, Approval flow, Digital cert PDF, Status track, Notifications, Admin panel, Export, Postman API

Problem 354: Land Records & Pahani Portal

Domain: Meeseva (Gov Services)

Difficulty: ★★ Intermediate

Description:

View and download land records (Pahani/RoR) by survey number, request mutation, and track encumbrance certificate applications.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Survey search, Pahani display, Mutation request, EC application, Status track, PDF download, History, Notifications, Admin panel, Postman API

Problem 355: Property Tax Payment Portal

Domain: Meeseva (Gov Services)

Difficulty: ★★★ Advanced

Description:

Calculate, pay, and download receipts for property tax with self-assessment, exemption check, arrears view, and OC/CC request.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Property search, Tax calc, Exemption check, Arrears display, Payment form, Receipt PDF, History, Notifications, Admin panel, Postman API

Problem 356: Water & Sewerage Connection Application

Domain: Meeseva (Gov Services)

Difficulty: ★ Beginner

Description:

Apply for new water/sewerage connections with property details, site inspection scheduling, feasibility display, and connection status.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Application form, Property input, Inspection schedule, Feasibility display, Status track, Connection cert PDF, Notifications, Admin panel, Export, Postman API

Problem 357: Trade License Application Portal

Domain: Meeseva (Gov Services)

Difficulty: ★★ Intermediate

Description:

Apply for new or renewal trade licenses with business details, NOC uploads, fire/health department routing, and certificate download.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Application form, Business details, NOC upload, Dept routing, Status track, License PDF, Renewal reminder, Notifications, Admin panel, Export

Problem 358: Old Age & Widow Pension Application

Domain: Meeseva (Gov Services)

Difficulty: ★ Beginner

Description:

Apply for government pension schemes with eligibility check, bank details, document upload, verification, and payment status.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Eligibility check, Application form, Doc upload, Bank details, Verification step, Payment status, Status track, Notifications, Admin panel, Export

Problem 359: Disability Certificate Application

Domain: Meeseva (Gov Services)

Difficulty: ★ ★ Intermediate

Description:

Apply for disability certificates with medical assessment details, hospital certification upload, approval workflow, and digital issuance.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Application form, Medical cert upload, Disability type, Approval flow, Certificate PDF, Status track, Notifications, Admin panel, Export, Postman API

Problem 360: Encumbrance Certificate Application

Domain: Meeseva (Gov Services)

Difficulty: ★ ★ ★ Advanced

Description:

Apply for EC with property registration details, period selection, payment, processing, and certified download.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Property input, Period select, Payment mock, Processing status, EC PDF download, History, Notifications, Admin panel, Export, Postman API

Problem 361: Voter ID Services Portal

Domain: Meeseva (Gov Services)

Difficulty: ★ ★ Intermediate

Description:

Apply for new voter ID, corrections, address change, and duplicate card requests with form 6/7/8 mock and status tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Form select (6/7/8), Application input, Doc upload, Status track, Card display, Booth info, History, Notifications, Admin panel, Export

Problem 362: Ration Card Services Portal

Domain: Meeseva (Gov Services)

Difficulty: ★ ★ ★ Advanced

Description:

Apply for new ration card, modifications, member addition/deletion, surrender, and smart card linking with status tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Application form, Member manage, Doc upload, Surrender flow, Smart card link, Status track, Notifications, Admin panel, Export, Postman API

Problem 363: Driving Licence Application Portal

Domain: Meeseva (Gov Services)

Difficulty: ★ Beginner

Description:

Apply for learner's licence, DL test scheduling, original DL, and renewal with eligibility check, fee payment, and slot booking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Application form, Eligibility check, Test scheduler, Fee payment mock, Slot booking, Status track, Certificate PDF, Notifications, Admin panel, Export

Problem 364: Vehicle Registration & Ownership Transfer

Domain: Meeseva (Gov Services)

Difficulty: ★★ Intermediate

Description:

Apply for vehicle registration, RC copy, fitness certificate, NOC, and ownership transfer with documents and fee payment.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Vehicle form, Doc upload, Fee payment mock, NOC request, Transfer form, Status track, RC PDF, Notifications, Admin panel, Postman API

Problem 365: Passport Application Assistance Portal

Domain: Meeseva (Gov Services)

Difficulty: ★★★ Advanced

Description:

Pre-fill and guide passport form submission, book PSK appointments, track application, and manage police verification status.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Form guide, Appointment book, PSK slots, Status track, Police verify status, Document checklist, Notifications, Admin panel, Export, Postman API

Problem 366: Scholarship Application Portal (SC/ST/BC/OBC)

Domain: Meeseva (Gov Services)

Difficulty: ★ Beginner

Description:

Apply for government scholarships with category-wise eligibility check, income proof, academic records, bank details, and disbursement status.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Eligibility check, Application form, Doc upload, Bank details, Status track, Disbursement display, History, Notifications, Admin panel, Export

Problem 367: Social Security Pension Dashboard

Domain: Meeseva (Gov Services)

Difficulty: ★★ Intermediate

Description:

Manage social security pension beneficiaries — NSAP, AABY, Aasara — with enrolment, payment status, and grievance link.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Beneficiary list, Enrolment form, Payment status, Grievance link, Update details, History, Notifications, Export, Admin panel, Postman API

Problem 368: Electricity Bill Payment & New Connection

Domain: Meeseva (Gov Services)

Difficulty: ★ Beginner

Description:

Pay electricity bills, apply for new connections, request load enhancements, and submit meter reading with DISCOM integration mock.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Bill payment form, New connection form, Load enhance, Meter reading, Payment receipt PDF, Status track, History, Notifications, Admin panel, Postman API

Problem 369: Meeseva Kiosk Operator Dashboard

Domain: Meeseva (Gov Services)

Difficulty: ★★ Intermediate

Description:

Kiosk operator UI to submit, track, and manage citizen applications across all Meeseva services with queue management.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Service select, Application form, Queue display, Status update, Print cert, Payment record, History, Export, Notifications, Admin panel

Problem 370: Right to Information (RTI) Portal

Domain: Meeseva (Gov Services)

Difficulty: ★★★ Advanced

Description:

File RTI applications online, track first appeals, second appeals, and receive information responses with fee payment.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

RTI form, Dept select, Fee payment mock, Status track, Response view, Appeal form, History, Notifications, Admin panel, Export

Problem 371: Telangana Building Permissions Portal (GHMC)

Domain: Meeseva (Gov Services)

Difficulty: ★ ★ Intermediate

Description:

Apply for building plan approval, occupancy certificate, completion certificate, and revisions with drawing upload and inspection scheduling.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Application form, Drawing upload, Inspection schedule, Revision submit, Status track, Certificate PDF, Notifications, Admin panel, Export, Postman API

Problem 372: Integrated Grievance Redressal System (IGRS)

Domain: Meeseva (Gov Services)

Difficulty: ★ ★ ★ Advanced

Description:

Register and track public grievances to government departments with priority routing, escalation, and resolution SLA tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Grievance form, Dept route, Priority badge, Status track, Escalation flow, Resolution note, SLA display, History, Export, Admin panel

Problem 373: Meeseva Document Verification Portal

Domain: Meeseva (Gov Services)

Difficulty: ★ Beginner

Description:

Verify authenticity of Meeseva-issued certificates by QR code or reference number with digital signature validation.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

QR scan input, Reference number, Verification result, Digital sig display, Certificate preview, History, Admin panel, Export, Notifications, Postman API

Problem 374: Village/Ward Secretariat Services

Domain: Meeseva (Gov Services)

Difficulty: ★★ Intermediate

Description:

Access village/ward secretariat services — applications, certificates, complaints, welfare scheme enrolment — from a unified portal.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Service catalogue, Application form, Status track, Scheme enrol, Complaint form, Notifications, History, Export, Admin panel, Postman API

Problem 375: PM Kisan & Agriculture Subsidy Tracker

Domain: Meeseva (Gov Services)

Difficulty: ★★★ Advanced

Description:

Track PM Kisan payments, apply for agriculture subsidies, input scheme, crop insurance, and view disbursement history.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Payment tracker, Subsidy form, Input scheme, Crop insurance, Disbursement history, Bank verify, Notifications, Admin panel, Export, Postman API

Problem 376: Weaver/Artisan Registration & Benefit Portal

Domain: Meeseva (Gov Services)

Difficulty: ★ Beginner

Description:

Register weavers and artisans for government schemes, tool subsidies, health insurance, and training with certificate issuance.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Registration form, Scheme select, Doc upload, Tool subsidy, Health ins, Training link, Certificate PDF, History, Notifications, Admin panel

Problem 377: Fire NOC Application Portal

Domain: Meeseva (Gov Services)

Difficulty: ★★ Intermediate

Description:

Apply for fire safety NOC for commercial premises with building details, floor plan upload, inspection scheduling, and certificate download.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Application form, Floor plan upload, Inspection schedule, Status track, Fire NOC PDF, Renewal reminder, Notifications, Admin panel, Export, Postman API

Problem 378: Meeseva Fee Payment & Receipt System

Domain: Meeseva (Gov Services)

Difficulty: ★ Beginner

Description:

Centralised fee payment for all Meeseva services with multiple payment modes, challan generation, refund, and audit trail.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Service fee list, Payment form, Challan PDF, Refund request, Audit trail, History, Notifications, Export, Admin panel, Postman API

Problem 379: Marriage Certificate Application Portal

Domain: Meeseva (Gov Services)

Difficulty: ★★ Intermediate

Description:

Apply for marriage certificates with wedding proof documents, joint application, witness details, and digital certificate issuance.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Joint application form, Doc upload, Witness input, Payment mock, Certificate PDF, Status track, Notifications, Admin panel, Export, Postman API

Problem 380: Meeseva API Integration Testing Dashboard

Domain: Meeseva (Gov Services)

Difficulty: ★ ★ ★ Advanced

Description:

Developer and operator portal to test Meeseva REST APIs — certificate issuance, status check, payment — using Postman-style tool.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Request builder, Auth header, Service endpoint select, Response viewer, Test suite, Error log, History, Export, Notifications, Docs panel

Problem 381: RERA Property Complaint Portal (Telangana)

Domain: Meeseva (Gov Services)

Difficulty: ★ ★ Intermediate

Description:

File complaints against builders/agents under RERA with project details, grievance documentation, hearing scheduling, and order tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Complaint form, Project search, Doc upload, Hearing schedule, Status track, Order display, History, Notifications, Admin panel, Export

Problem 382: Citizen Feedback on Government Services

Domain: Meeseva (Gov Services)

Difficulty: ★ ★ ★ Advanced

Description:

Rate and review Meeseva and government service experiences with category ratings, suggestions, and anonymity option.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Feedback form, Service rating, Category score, Suggestion input, Anonymous toggle, Analytics display, History, Export, Notifications, Admin panel

Problem 383: Women & Child Welfare Scheme Portal

Domain: Meeseva (Gov Services)

Difficulty: ★ Beginner

Description:

Apply for schemes — Arogya Lakshmi, KCR Kits, Kalyana Lakshmi, Shaadi Mubarak — with eligibility check and beneficiary status.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Scheme catalogue, Eligibility check, Application form, Doc upload, Status track, Payment display, History, Notifications, Admin panel, Export

Problem 384: Employee Provident Fund (EPF) Services Portal

Domain: Meeseva (Gov Services)

Difficulty: ★ ★ Intermediate

Description:

Check EPF balance, claim withdrawals, update KYC, transfer accounts, and track claim status via EPFO integration mock.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Balance check, Withdrawal form, KYC update, Transfer form, Claim status, History, Notifications, Export, Admin panel, Postman API

Problem 385: SC/ST Corporation Loan Application Portal

Domain: Meeseva (Gov Services)

Difficulty: ★ ★ ★ Advanced

Description:

Apply for financial assistance from SC/ST corporations with scheme selection, income proof, guarantor details, and disbursement tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Scheme select, Application form, Income proof, Guarantor input, Status track, Disbursement display, History, Notifications, Admin panel, Export

Problem 386: Municipal Solid Waste Complaint Portal

Domain: Meeseva (Gov Services)

Difficulty: ★ Beginner

Description:

Report solid waste issues — uncollected garbage, illegal dumping, overflowing bins — with location, photo, and resolution tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Complaint form, Location input, Photo upload, Category select, Status track, Resolution note, History, Notifications, Admin panel, Export

Problem 387: Land Acquisition & Compensation Tracker

Domain: Meeseva (Gov Services)

Difficulty: ★★ Intermediate

Description:

Track government land acquisition cases, compensation awards, objection filing, and payment status for affected landowners.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Case search, Acquisition details, Objection form, Compensation display, Payment status, History, Notifications, Admin panel, Export, Postman API

Problem 388: e-District Courts & Legal Aid Portal

Domain: Meeseva (Gov Services)

Difficulty: ★ Beginner

Description:

Apply for legal aid services, view court dates, pay court fees, request certified copies of court documents, and track case status.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Application form, Court fee payment, Case status track, Legal aid apply, Doc request, Certified copy download, History, Notifications, Admin panel, Export

Problem 389: Meeseva Statistical & MIS Dashboard

Domain: Meeseva (Gov Services)

Difficulty: ★★ Intermediate

Description:

Real-time MIS dashboard for Meeseva administrators showing applications processed, pending, rejected, revenue collected, and operator performance.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

KPI dashboard, Application chart, Pending queue, Revenue display, Operator performance, Date filter, Export, History, Notifications, Admin panel

Problem 390: Heritage & Tourism Registration Portal (AP/TS)

Domain: Meeseva (Gov Services)

Difficulty: ★ ★ ★ Advanced

Description:

Register tourism establishments, apply for heritage site visitor passes, book guided tours, and access conservation programme info.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Establishment form, Pass application, Tour booking, Conservation info, Certificate PDF, Status track, History, Notifications, Admin panel, Export

Problem 391: AP/TS e-Procurement Portal UI

Domain: Meeseva (Gov Services)

Difficulty: ★ ★ Intermediate

Description:

Browse government tenders, download bid documents, submit bids, track evaluation, and view award notices for public procurement.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Tender catalogue, Bid download, Submission form, Evaluation status, Award notice, History, Notifications, Admin panel, Export, Postman API

Problem 392: NREGA Job Card & Work Application Portal

Domain: Meeseva (Gov Services)

Difficulty: ★ ★ ★ Advanced

Description:

Apply for NREGA job cards, request work, track daily attendance, wages, and muster rolls for rural employment.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Job card form, Work request, Attendance log, Wage display, Muster roll, Status track, History, Notifications, Admin panel, Export

Problem 393: District Collector Petition Portal

Domain: Meeseva (Gov Services)

Difficulty: ★ Beginner

Description:

Submit petitions to the district collector office with category, urgency flag, document attachment, and disposal status tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Petition form, Category select, Urgency flag, Doc attach, Status track, Response view, History, Notifications, Admin panel, Export

Problem 394: Flood/Disaster Relief Application Portal

Domain: Meeseva (Gov Services)

Difficulty: ★ ★ Intermediate

Description:

Apply for disaster relief assistance with damage assessment photos, family details, bank account, and relief disbursement tracking.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Application form, Damage photos, Family input, Bank details, Assessment review, Status track, Disbursement display, History, Notifications, Admin panel

Problem 395: Smart Governance Citizen App Dashboard

Domain: Meeseva (Gov Services)

Difficulty: ★ ★ ★ Advanced

Description:

Unified citizen dashboard aggregating all Meeseva services, pending applications, dues, scheme benefits, and government notifications.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Service hub, Pending apps, Dues display, Scheme benefits, Govt notifications, Profile, Quick actions, History, Export, Admin panel

Problem 396: Meeseva Helpdesk & Escalation Portal

Domain: Meeseva (Gov Services)

Difficulty: ★ Beginner

Description:

Citizen and operator helpdesk for Meeseva service issues with ticket submission, call back request, escalation, and resolution.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Ticket form, Callback request, Category select, Status track, Escalation flow, Resolution note, History, Notifications, Export, Admin panel

Problem 397: Digital Locker (DigiLocker) Integration UI

Domain: Meeseva (Gov Services)

Difficulty: ★ ★ Intermediate

Description:

Upload, organise, and access government-issued documents from DigiLocker mock with service link, QR share, and validity display.

Tools & Technologies:

HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, npm, Postman (API Testing), MySQL (optional), REST APIs, Express.js, Axios, React Router, Context API / Redux, LocalStorage, Bootstrap / Tailwind CSS

Modules to Implement (min 10):

Document list, Upload form, DigiLocker mock, QR share, Validity display, Service link, History, Notifications, Admin panel, Export

Appendix: Implementation Guidelines

Minimum Module Requirements (per problem)

Every problem in this document must implement at minimum 10 distinct modules. The following standard modules apply across all domains:

- Module 1 — Authentication & Authorization: User registration, login, JWT/session mock, role-based access
- Module 2 — Dashboard / Home Screen: Role-appropriate landing page with KPI cards and quick actions
- Module 3 — Core Feature A: Primary domain functionality (search, booking, application, etc.)
- Module 4 — Core Feature B: Secondary domain functionality (tracking, management, analytics, etc.)
- Module 5 — Form Handling: Validated multi-field forms with error handling and success states
- Module 6 — Data Display: Tables, lists, or card grids with sorting, filtering, and pagination
- Module 7 — Notifications & Alerts: Toast notifications, modal alerts, and in-app notification centre
- Module 8 — File Handling: Document upload, preview, download PDF receipts or reports
- Module 9 — API Integration: REST API calls via Axios with loading, success, and error states tested in Postman
- Module 10 — Admin Panel: Administrative controls for data management, user management, and reporting
- Module 11 (Optional) — Analytics & Charts: Data visualisation using Chart.js or Recharts
- Module 12 (Optional) — MySQL Integration: Persistent data storage with Express.js backend and MySQL

Postman Testing Requirements

Each project must include a Postman collection with test cases for: (a) all API endpoints created, (b) authentication flows, (c) CRUD operations, (d) error response handling, and (e) environment variable usage for base URLs and tokens.

Evaluation Criteria

- UI/UX Quality — Responsive design, accessibility, visual hierarchy, and intuitive navigation
- Code Quality — Clean, well-structured React components, separation of concerns, and reusable code
- API Design — RESTful endpoint design, proper HTTP methods, status codes, and Postman documentation
- Feature Completeness — Minimum 10 modules implemented and functional
- Innovation — Additional features, animations, or novel solutions beyond the base requirements
- Presentation — Demo quality, documentation, and ability to explain architectural decisions

*****ALL THE BEST *****